Designing for all by designing for Niche

BMVSS- Jaipur Foot

Human centered design & Experience Design - PG year 1

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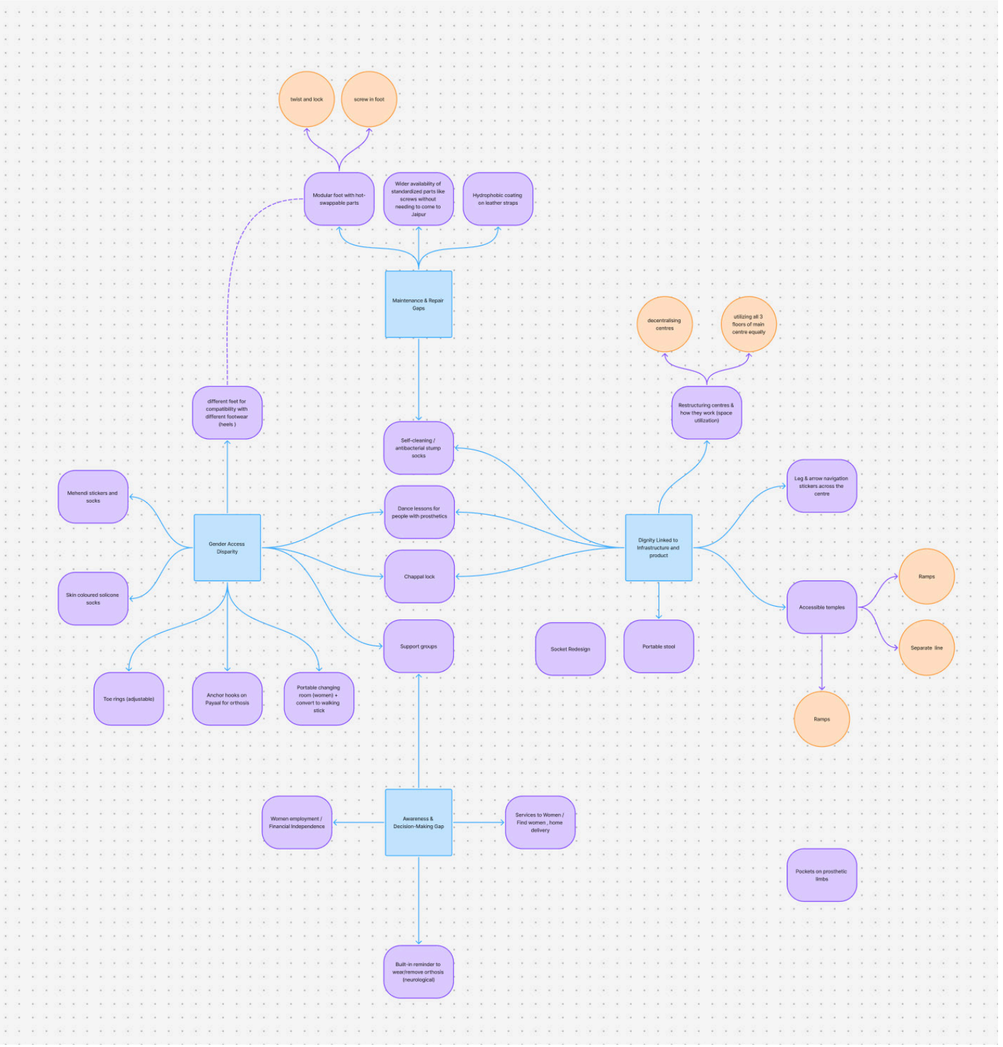
A2

How to make services available for women and children?

How to enhance the experience of women with prosthetics (personalization)?

Phase 1 - Ideas (Exploring)

[Link to Ideas Phase 1 :]



After synthesizing primary visits and secondary research, we began by putting down all possible ideas without filtering. This phase was intentionally exploratory capturing everything from quick fixes to system-level thoughts, across comfort, mobility, accessibility, and social experience, repair, Gender Access Disparity, infrastructure, awareness.

Once we had a wide pool of ideas, we started categorising them into broader themes i. e Dignity Linked to Infrastructure and product, Maintenance & Repair Gaps, Gender Access Disparity, Awareness & Decision-Making Gap. These were inspires by our contexts as well.

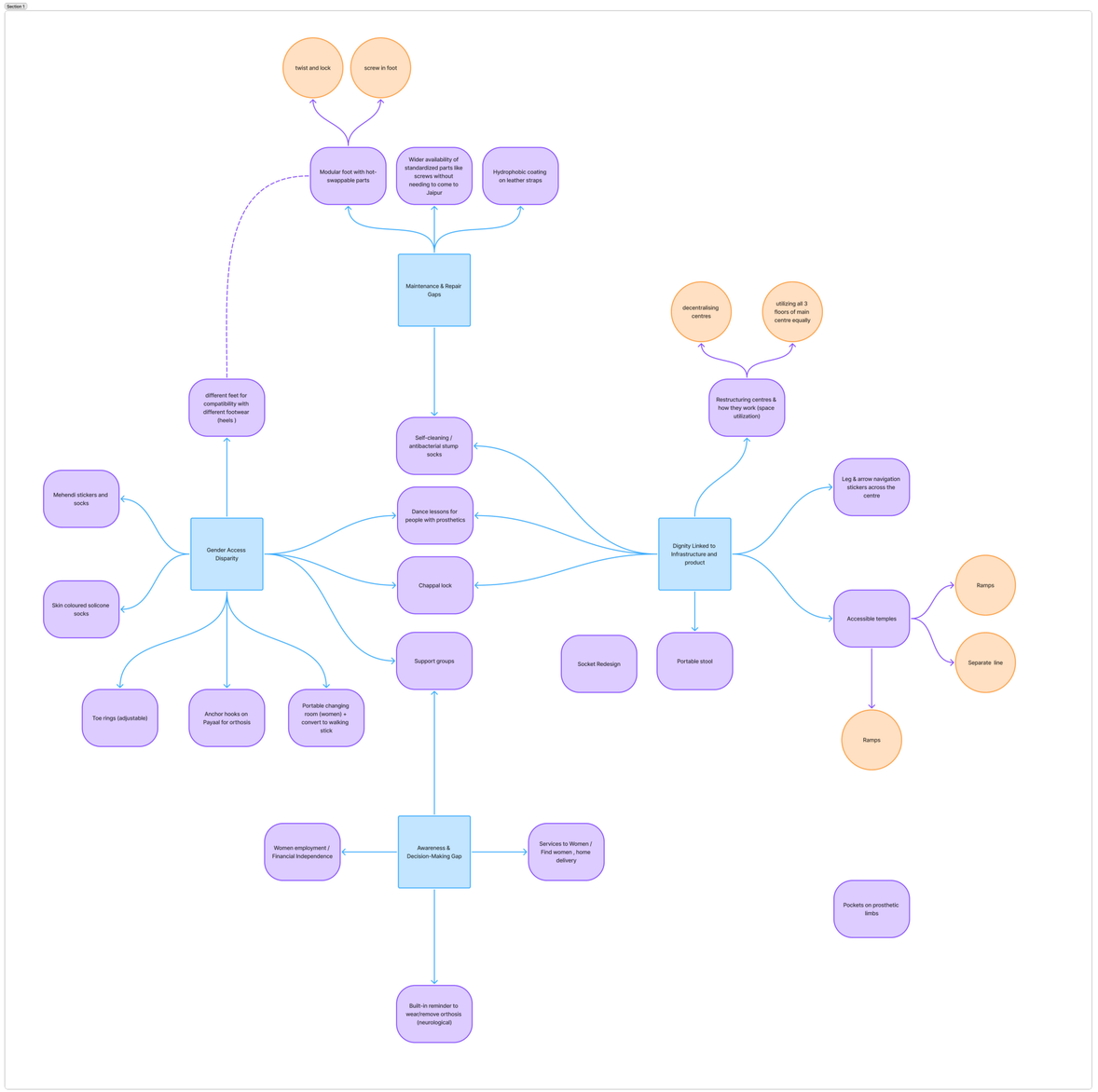
The we started mapping these ideas visually, looking at overlaps, relationships, and potential directions. Some ideas connected across multiple categories, revealing deeper opportunity areas rather than isolated solutions.

This was about exploring , enjoying , and trusting the process.

[Link to Ideas Phase 1 :]

Phase 1 - Ideas (Exploring)

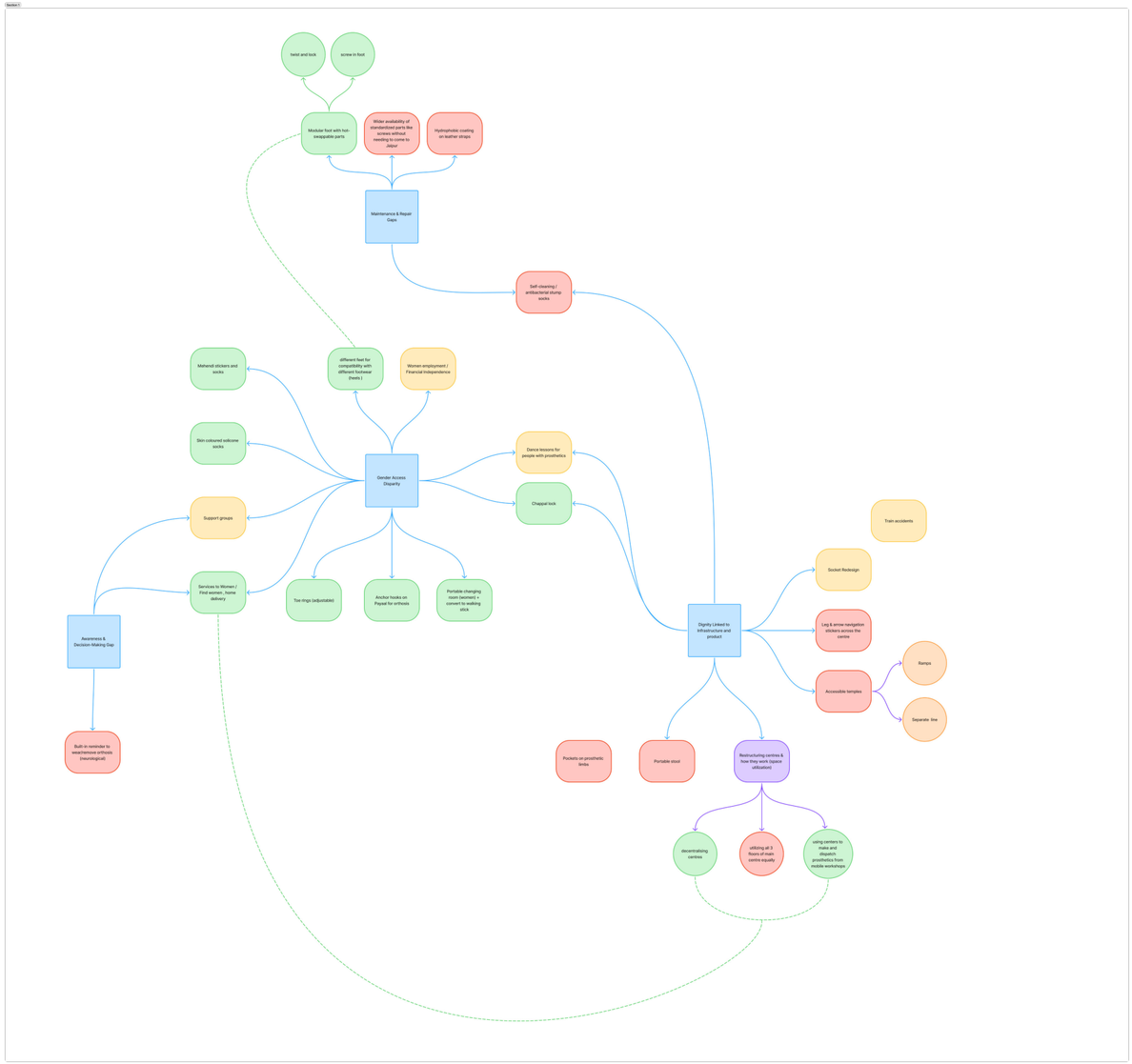
[Link to Ideas Phase 1 :]



[Link to Ideas Phase 1 :]

Phase 2- Ideas (Categorization)

Link to Ideas Phase 2 :



Building on the initial pool of ideas, in Phase 2 even though we did come up with some concepts but it was more about making decisions. The primary focus was to categorise and filter ideas based on feasibility within the given timeframe (2–3 weeks).

Link to Ideas Phase 2 :

Phase 2- Ideas (Categorization)

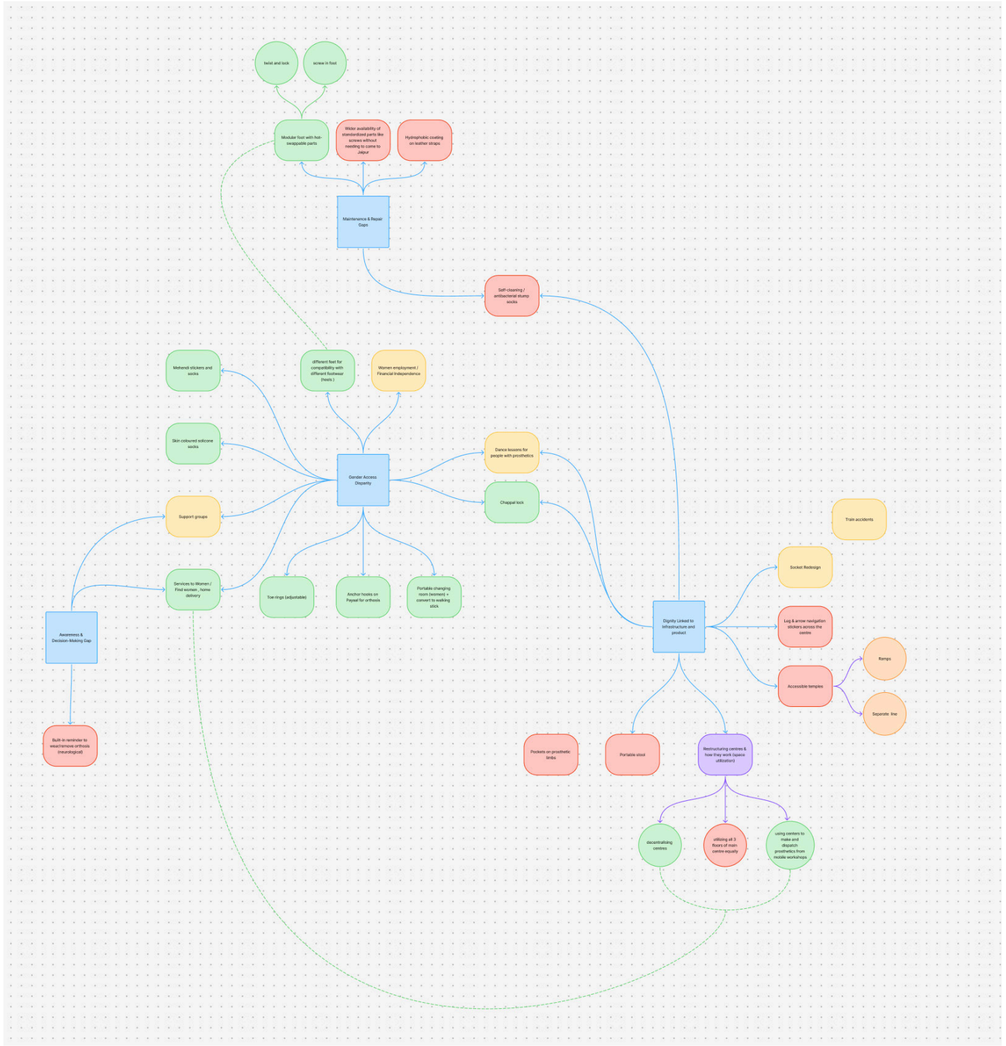
[Link to Ideas Phase 2 :]

We organised the ideas using a colour-coded system to clearly distinguish between levels of execution:

- Red – Future scope
- Yellow – Might be able to do
- Green – Feasible
- Blue – Categories

This made sure we did practical decision-making, ensuring that the selected ideas were not only relevant but also realistic to prototype and test within the course constraints.

Rather than pursuing everything, it was about narrowing down to scope of the project that could be meaningfully be developed further.



[Link to Ideas Phase 2 :]

Phase 3- Ideas (Execution Tools)

[Link to Ideas Phase 3 :]

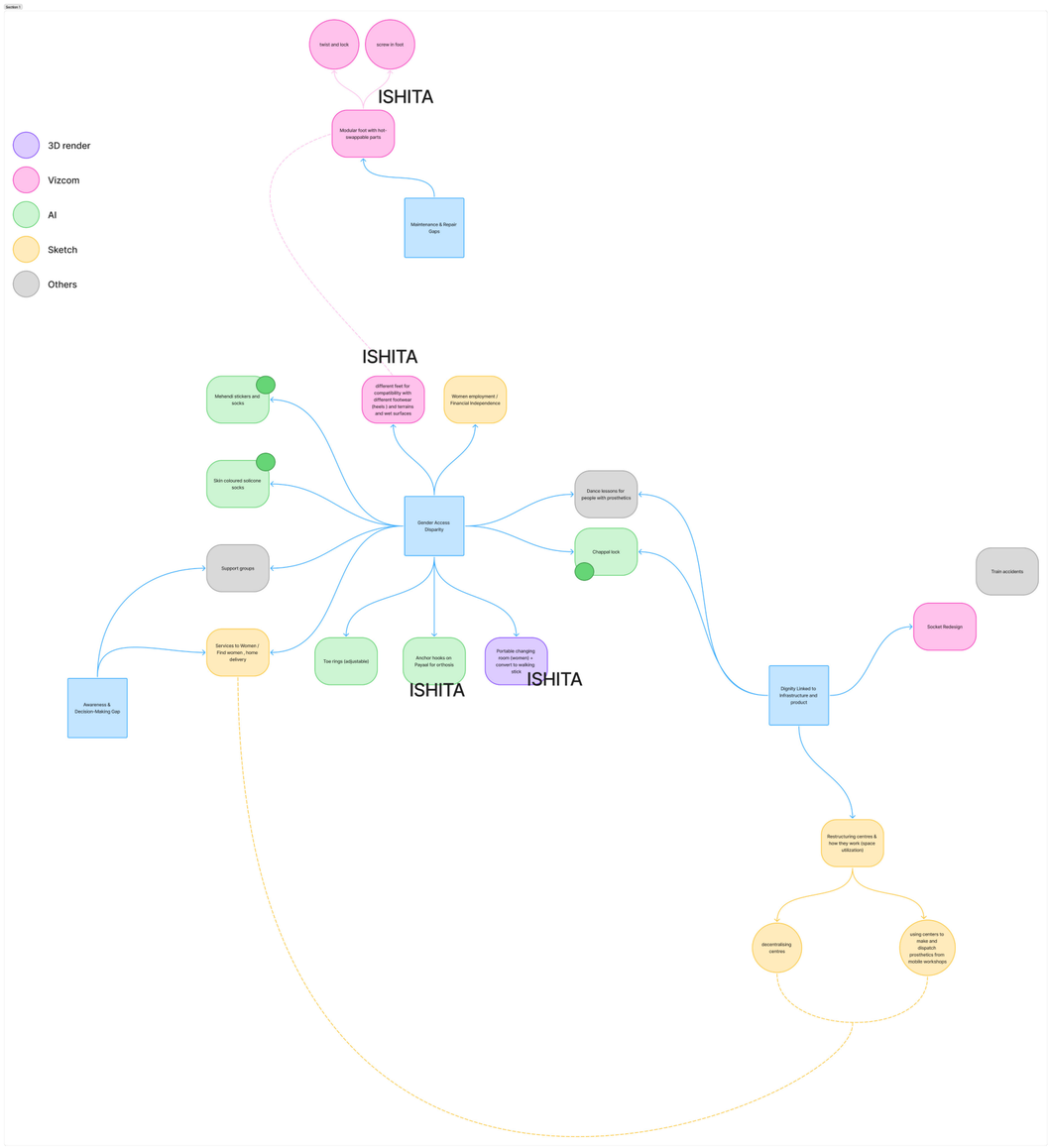
3D render

Vizcom

AI

Sketch

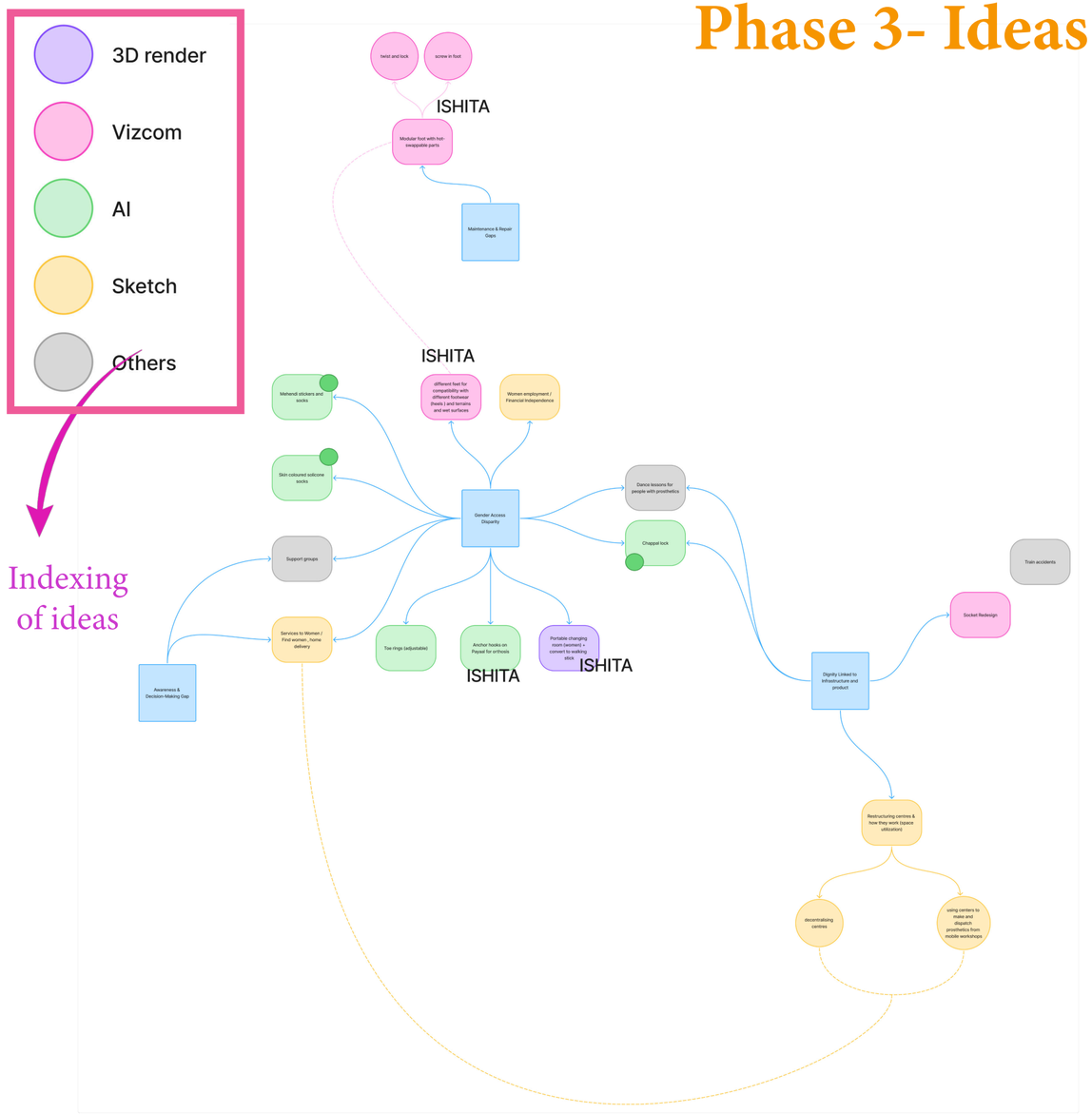
Others



[Link to Ideas Phase 3 :]

Phase 3- Ideas (Execution Tools)

[Link to Ideas Phase 3 :]



In Phase 3, we started thinking how these ideas could be executed. To streamline this, we categorisation ideas based on the type of tools and software required for their development and representation.

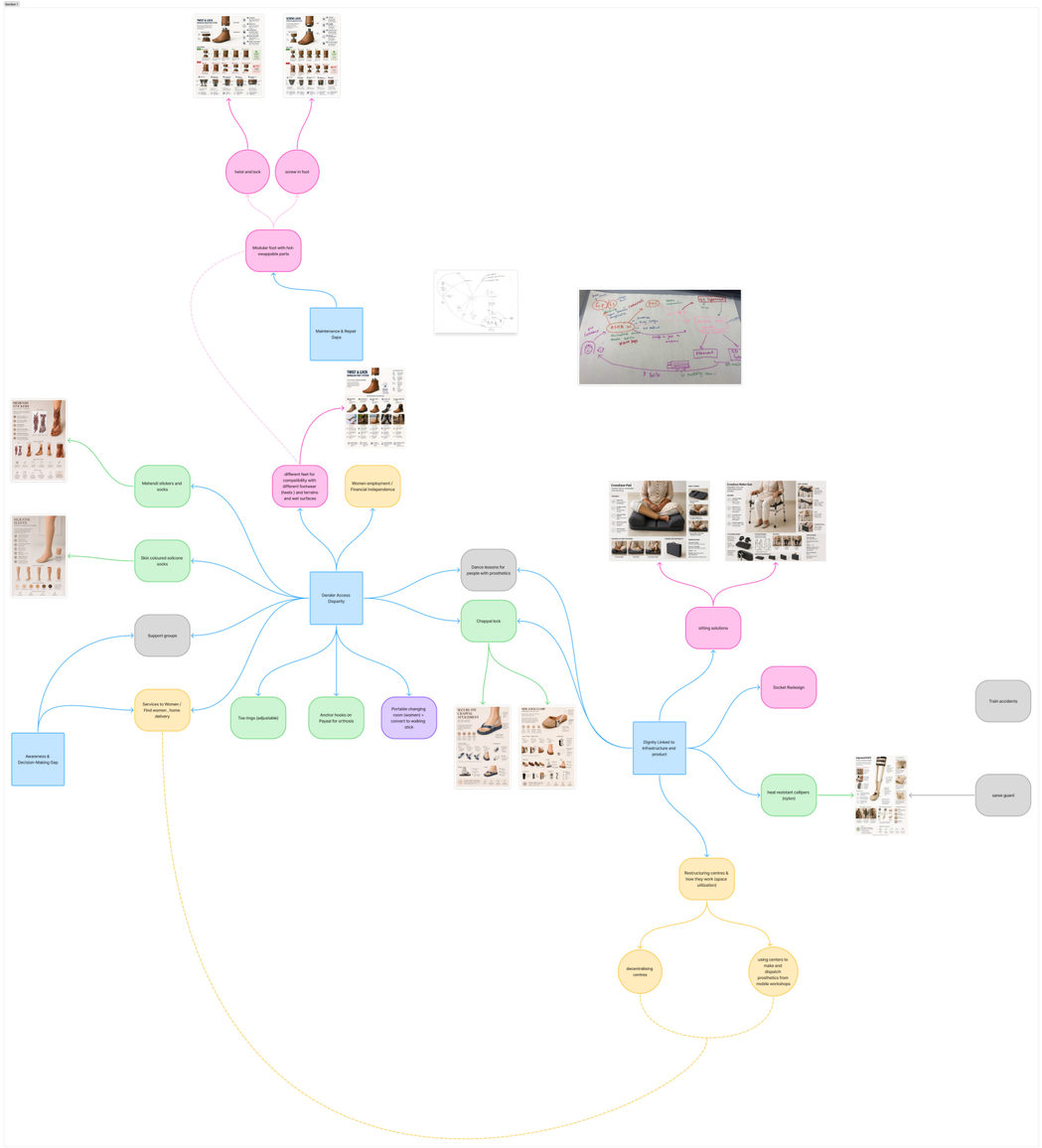
The categorisation marked at right top corner.

This was for us to help imagine how the idea would look like in real world scenario and also would help us understand the possible challenges.

[Link to Ideas Phase 3 :]

Phase 4 - Ideas (Generated)

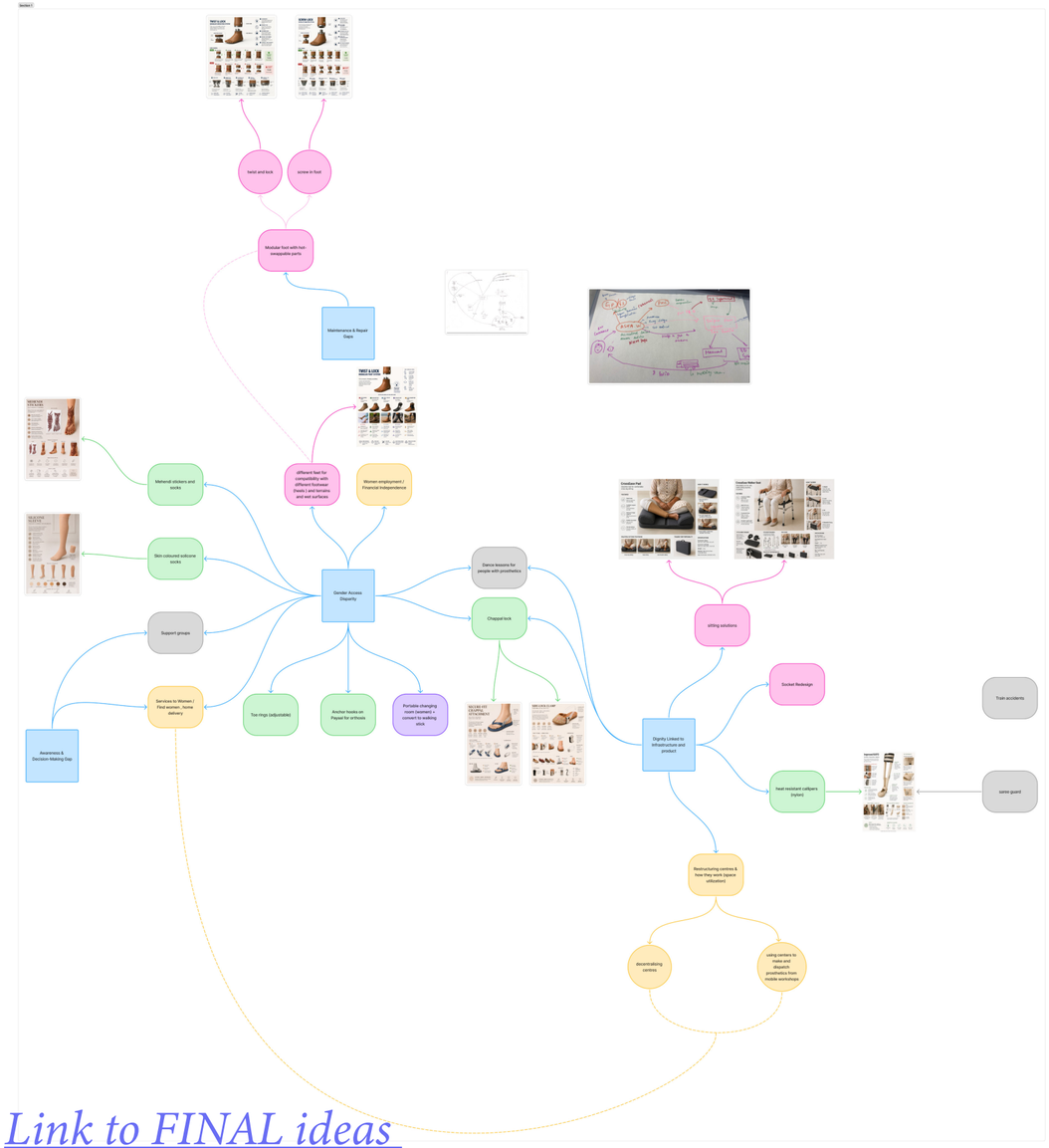
[Link to Ideas Phase 4:]



[Link to Ideas Phase 4:]

Phase 4 - Ideas (Generated)

[Link to FINAL ideas](#)



[Link to FINAL ideas](#)

In the final stage, we continued to generate a wide range of ideas, exploring multiple directions based on our research and insights. However, due to time constraints and the need to go deeper rather than broader, we decided to prototype only two selected ideas.

These ideas were chosen based on:

- Relevance to key pain points
- Feasibility within the given timeframe
- Potential for real-world impact

The selected ideas :

1. Footwear Lock System

2. Home-Based Prosthetic Service Model

(more in prototype slides)

Link to FINAL ideas

SHORTLISTED IDEAS

Link to FINAL ideas

Link to Ideas Phase 4:

Opportunity area -1

"Prosthetics today are largely designed as medical devices, focusing on function and mobility, while overlooking the cultural, emotional, and identity-related needs of users especially women. In many contexts, appearance, self-expression, and social acceptance play a significant role in how individuals relate to their bodies and participate in society. This creates an opportunity to reimagine prosthetics not just as functional aids, but as extensions of personal and cultural identity."

Evidence from interviews - During our field interactions and interviews, a recurring pattern emerged where several women expressed a desire to participate in cultural practices like applying mehendi, especially during weddings and festivals, but felt limited due to their prosthetics. While they often applied mehendi on their hands, many hesitated or avoided doing so on their prosthetic limbs, as it did not feel like a natural extension of their body.

Link to Ideas Phase 4:

Sol next page ->

Mehendi Stickers

Reference

Our first idea for which we used AI to conceptualize our ideas, we started off by using this image taken at BMVSS Jaipur as reference image

Iteration

next we kept on iterating on the render to get something that looks closer to what we require

Concept

Once the Idea was visualised, the next step was to conceptualize the product and create a visual infographic, after 3-4 iterations on the concept, this was our final version.

Link to Ideas Phase 4:

Opportunity area -2

"Prosthetic limbs are often visually distinct from the user's natural body, particularly in terms of skin tone, texture, and finish, which can make them stand out in social settings. For many women, this difference affects not just appearance but also comfort, confidence, and willingness to participate in everyday and cultural activities. The lack of personalization in prosthetics creates a gap where the device feels more like an external object rather than an extension of the body, impacting self-perception and social ease."

Evidence from interviews - During field interactions, several women expressed discomfort with how visibly different their prosthetic looked compared to their natural skin tone. Some mentioned feeling conscious in public spaces or social gatherings, often choosing clothing or behavior that concealed the prosthetic.

Link to Ideas Phase 4:

Sol next page ->

Silicone Sleeve

Reference

For this idea we started off by using one of the iterations for our mehendi sticker ideas where we swapped out the cloth sleeve in the reference for a silicone one.

Concept 1

We isolated the idea of the silicone sleeve from the iteration to create our first concept for this product.

Concept 2

We further iterated on the concept by making the silicone sleeve cover the foot area as well and added more details to the infographic such as a visual overview of just the product itself

Concept 3

For the final concept we added a wide variety of different skin tones to the product and a textured grip at the sole. but based on peer review of this concept the product might have some structural issues due to which we might go back to a shape closer to the first concept while maintaining the functionality of the skin tone options.

Link to Ideas Phase 4:

Opportunity area -3

"Prosthetic feet often lack the grip, flexibility, and toe engagement needed to securely hold everyday footwear like chappals or women sandals, jooti etc . This creates a significant challenge for users, especially women, as their commonly worn footwear tends to slip off while walking. The issue becomes more pronounced in wet or uneven conditions, limiting footwear choices and affecting stability, safety, and confidence in daily movement."

Evidence from interviews - During field observations and interactions, multiple users reported that their chappals frequently slipped off while walking, and they won't even notice , hence slipper would never be found. This was particularly noticeable in wet areas, where the lack of grip increased the risk of slipping. As a result, many women restricted themselves to limited footwear options or avoided certain environments altogether. These recurring experiences highlight a gap where prosthetic design does not align with everyday footwear practices, pointing towards the need for a more secure and adaptable footwear interface.

Link to Ideas Phase 4:

Sol next page ->

Chappal Lock

Initial render

Brainstorming

For my prosthetics and orthotics design project, many people (especially women) wanted to wear footwear such as chappals (slippers) with their prosthetics. But the Jaipur foot doesn't have a gap between the toes. One woman had manually cut the gap, but she still had the issue of the chappal falling off her foot without her realising it. Help me brainstorm ideas for this.

currently I like both "The Invisible Slingback" and "The Anklet Tether (Payal concept)"

⚠️ Where It Needs Strengthening

1. Failure Mode: What if the Clip Slips?

Right now, your system depends on:

> clip → chappal strap friction

But chappal straps:

- Stretch over time
- Vary in thickness
- Can be sweaty/slippery

👉 If the clip shifts position, your whole system fails.

Fix direction:

- Add a **secondary locking geometry**
  - Micro teeth inside clip
  - Rubberized inner lining
  - Or a slight "hook-under" the strap

Concept

We started off this idea by using AI tools to create an initial render for our idea that we can use as reference going forward.

To further flesh out our idea, we shared our problem statement with an LLM alongside the idea and the render so that we could get feedback on it and brainstorm further iterations based on it.

Our next step was to incorporate all of the feedback from the AI to further iterate on our concept and come up with a version of it that fixes all of the previous issues.

Link to Ideas Phase 4:

Opportunity area -4

"While the initial chappal lock concept addressed slipper-based footwear, it became evident that prosthetic users especially women engage with a wider variety of culturally relevant footwear such as juttis and mules. These types of footwear lack back support and rely on natural foot grip, which prosthetic feet cannot replicate. As a result, users face difficulty in wearing such footwear independently, limiting their choices and restricting participation in everyday and cultural contexts."

Evidence from interviews - Through observations and user interactions, it was noted that women often avoided footwear like juttis or backless shoes despite their cultural and aesthetic significance, due to instability and difficulty in wearing them.

Link to Ideas Phase 4:

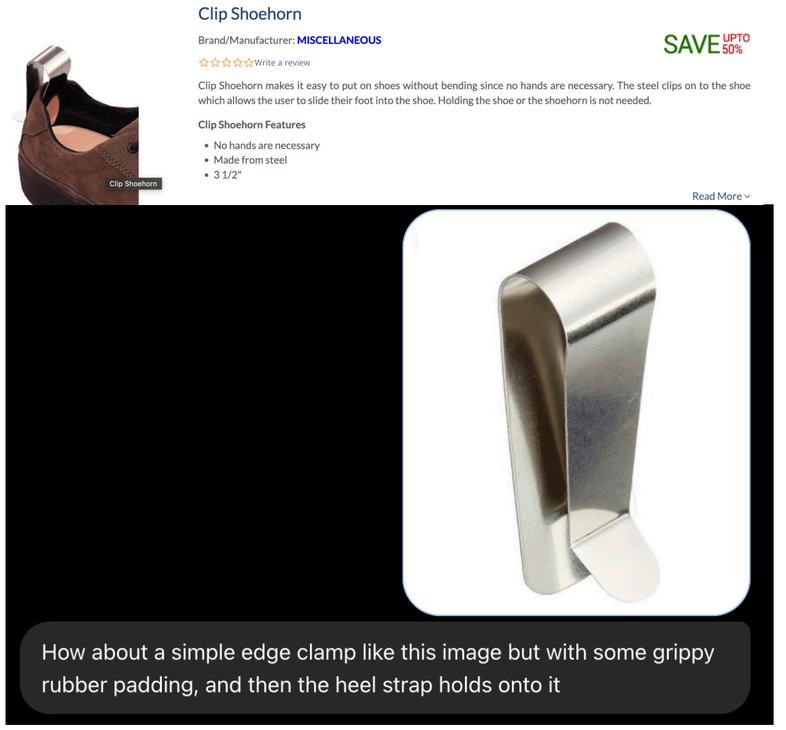
Sol next page ->

Universal Footwear Lock

Initial concept

After the chappal lock concept we decided to further branch out the idea and adapt it to work on a wider variety of footwear like mules and juttis. At first we gave the ai our current concept for the chappal lock, a reference image of a jutti and a rough sketch of our idea and asked it to adapt it for a wider range of foowear but the results were not satisfactory.

Research and further ideation



Facing issues with the AI, we did further research on existing accessibility solutions for footwear and came across the clip shoehorn, a product that helps people wear shoes hands free, we decided that this shape would be optimal for the clip design for this product and asked the AI to adapt it for our concept.

Concept

Incorporating the transparent back strap from the chappal lock design and the clip mechanism from the clip shoehorn we were finally able to create a concept for the mechanism that closely matched our vision for this product.

Link to Ideas Phase 4:

Opportunity area -5

"The existing prosthetic foot (such as the Jaipur Foot) is highly effective for navigating daily use but not might be very effective with varied terrains. However, it is primarily designed for functional use and does not accommodate the diverse footwear needs of users in terms of location and different foot wear , and especially for women who may wish to wear heels or different types of footwear for specific occasions. This highlights a gap where prosthetics lack adaptability to different lifestyle and aesthetic needs, limiting both convenience and personal expression."

Evidence from interviews - During field observations and interactions, it was noted that while users appreciated the durability in daily life and it lacked terrain adaptability of their existing prosthetic foot, they lacked flexibility in terms of footwear options. Some women expressed interest in wearing heels or occasion-based footwear but were unable to do so with their current setup. Additionally, switching or modifying prosthetic components required visits to centers like BMVSS, which involved time, travel, and dependency on technicians.

Link to Ideas Phase 4:

Sol next page ->

Modular Connection system

Initial ideation

Guided Entry (User-proof insertion)
Funnel-shaped socket entry
Asymmetrical pin layout → only fits one way

👉 Removes confusion + reduces failed attempts

2. Primary Lock: Bayonet Twist (~25° rotation)
2–3 metal pins on foot
L-shaped reinforced slots in limb

👉 This carries the load + weight

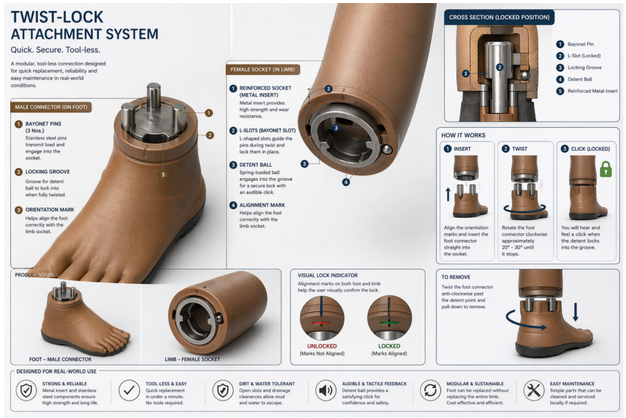
3. Secondary Lock: Click Detent (Critical)
Spring-loaded ball/pin inside socket
Engages with groove on foot connector

👉 Gives:

Audible click
Tactile confirmation
Resistance to accidental rotation

Being one of our most complex ideas, we started off the process for this concept by going back and forth with ai, brainstorming the different ways to implement a twist and lock mechanism to the pre existing jaipur foot to make it modular, solving a lot of current pain points with the prosthetic.

Concept 1

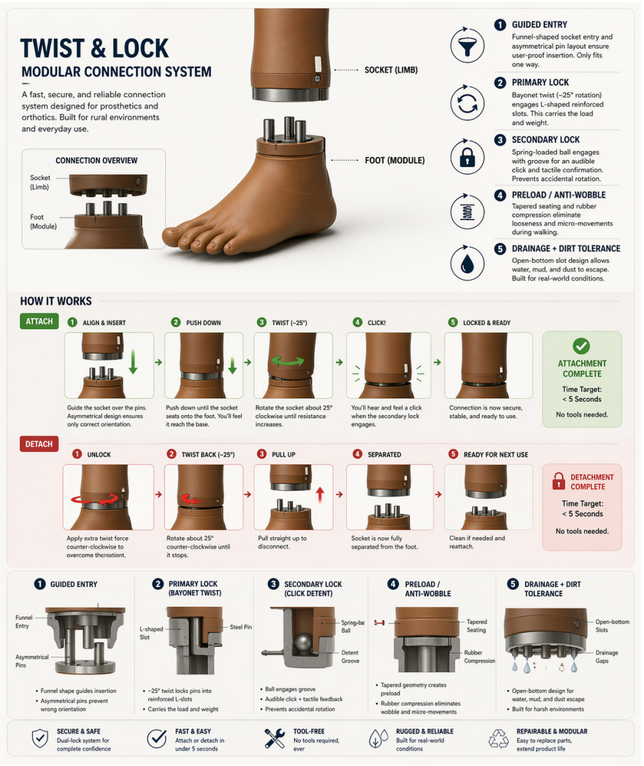


After we gave ai all of our current ideas this is the concept that we got, functionally covering all of our current ideas but having no visual consistenty due to lack of any reference images.

Product renders

To fix this issue we decided to use Vizcom, an ai powered product design tool to create renders of multiple variations and iterations of the product with visual consistency.

Final concept



We took our ideation and renders to start over for our final concept and after some minor iterations finally landed on this concept for our modular foot.

Modular Connection system- screw lock

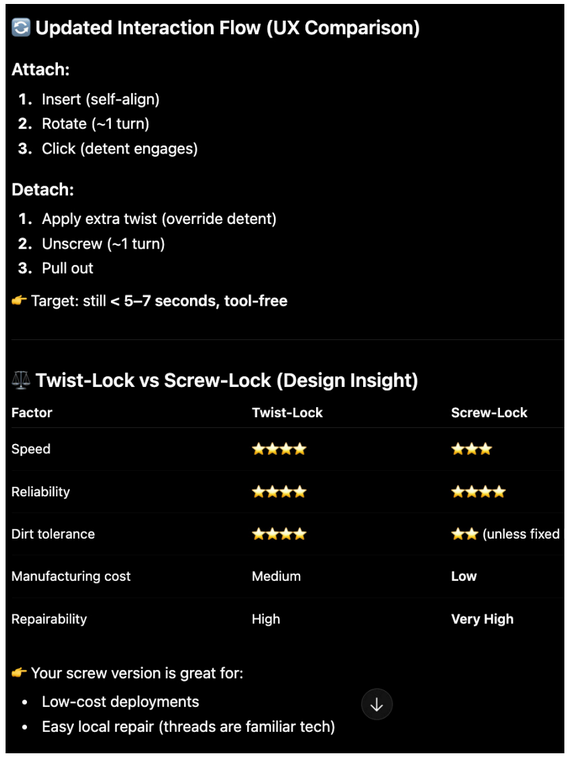
Idea inspiration

While talking to a farmer at BMVSS who has had a Jaipur foot since 1986, we came across a recurring pain points with the screws that connect the foot to the limb and how an older version of a prosthetic used to have a different system to screw the foot into the limb which inspired this alternative more cost-effective version of a modular system.

Product render

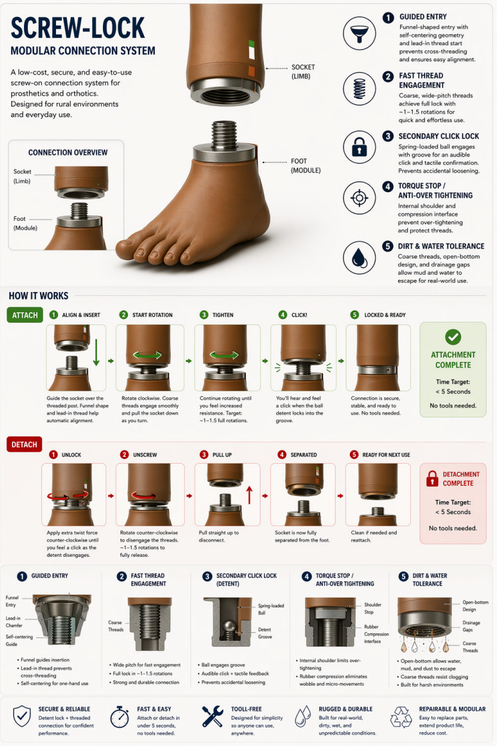
We used our last modular mechanism as reference and iterated on it further in vizcom to create a render of our product.

Idea Validation



Before turning the render into a concept we ran the idea and render through an LLM to figure out any functional issues with the idea and make minor changes to it to make sure that users don't run into issues

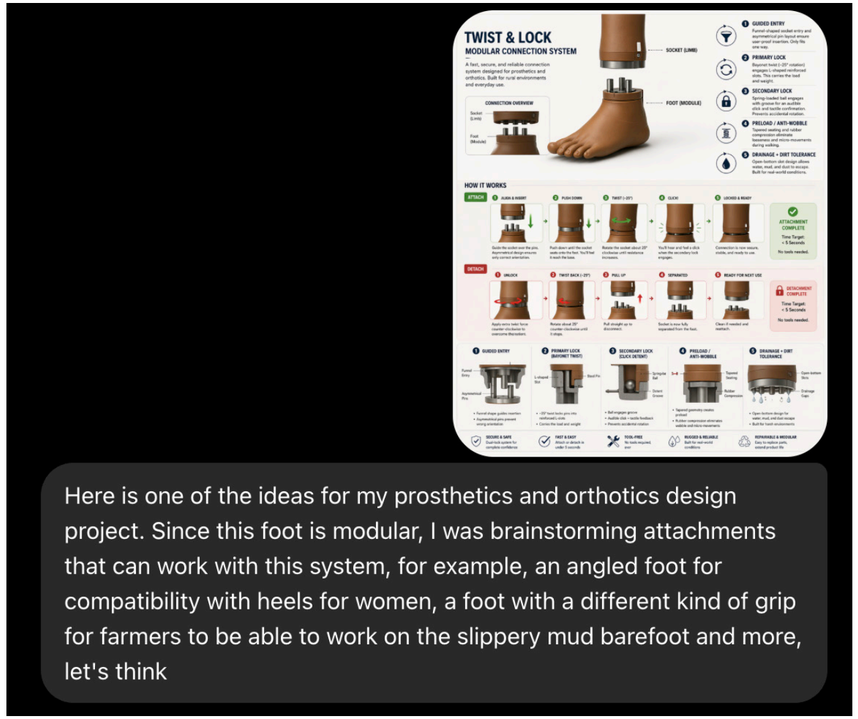
Final concept



We combined our render and the changes that we implemented based on the feedback from the LLM to come up with a final concept. Based on peer review we would have loked to further change and develop this concept given more time.

Modular Connection system- Modules

Baseline



We took our modular connection system and decided to create a wide variety of modules that can work in different scenarios a user might go through instead of a one size fits all solution. We already had ideas for modules such as an angled foot for women who want to hear heels and a more farmer oriented version of the foot with more grip and used ai for further ideation of modules.

Exploring possibilities

Leveraging the Ai, we looked into every single use case and the hypothetical module we can create for it.

Final concept

After looking into over a dozen hypothetical modules we decided to narrow it down to 5, our original ideas for the angled foot and the mud grip foot alongside 3 more modules, one for the sand, one for cycling and one for extended extensive use cases.

Link to Ideas Phase 4:

Opportunity area -6

"Orthotic devices such as KAFOs (calipers) are primarily designed for structural support and mobility, but often overlook the everyday comfort and cultural usability needs of women. For users with conditions like polio, prolonged use of metal orthoses can lead to heat discomfort, skin irritation, and friction, especially in hot climates. Additionally, the exposed structure of the device can snag onto clothing like sarees, creating inconvenience and limiting ease of movement. Existing fastening systems like velcro can also cause discomfort and lack refinement for long-duration wear."

Evidence from interviews -Through observations and discussions, women using orthoses reported discomfort due to direct contact with heated metal rods, particularly during extended wear in warm environments. There were also instances where sarees or loose garments would get caught in the orthotic structure, interrupting movement and causing frustration. Furthermore, fastening mechanisms such as velcro were found to be uncomfortable against the skin would cause blister and were not always user-friendly.

Link to Ideas Phase 4:

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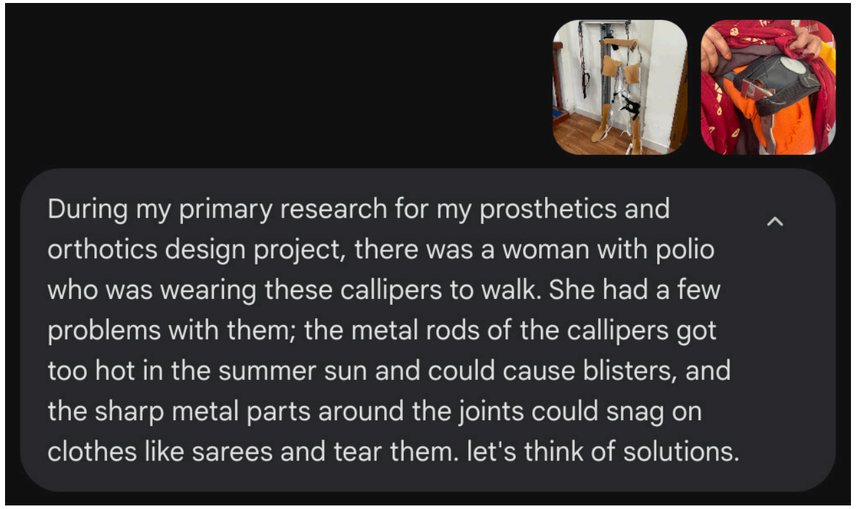
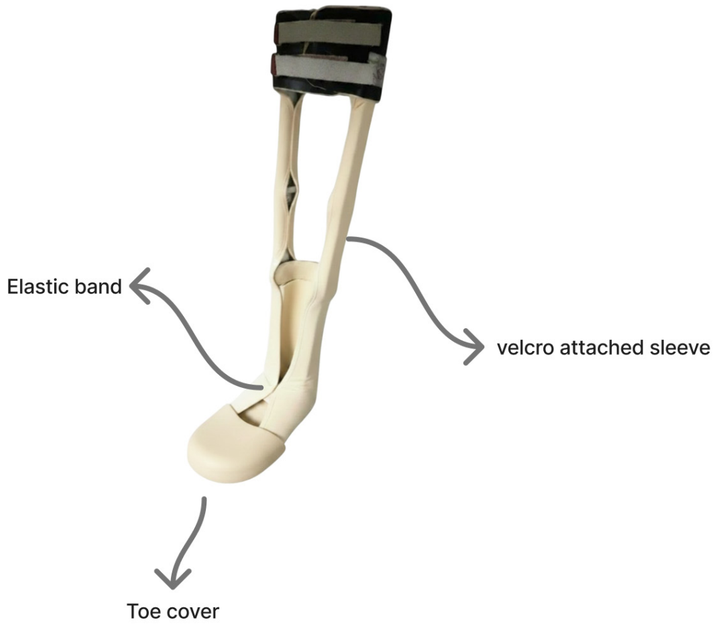
Improving the orthosis

Ideation



We started off with our problem statement regarding orthoses and the issues women with polio face with them on a daily basis. After brainstorming with AI around our problem statement we decided to combine 2 of our ideas into one, an insulating sleeve to prevent direct contact from the hot metal rods of the caliper and a protector/guard that would cover up the part of the orthoses that is the most likely to snag onto a saree. Aside from this consolidation of ideas we also had another idea that would replace the velcro on the current KAFO with a toe cover to provide additional comfort and prevent discomfort.

Render and labelling



After having our ideas in place we used vizcom to iterate upon the visual design of the solutions and create a render that included all 3 of the solutions that we initially came up with. once we had the visual design down we labelled the 3 solutions on the render for reference for any further iterations

Concept

We combined all 3 of our ideas with our render to creste our final concept.

Link to Ideas Phase 4:

Opportunity area -7

"Prosthetic devices are primarily designed for upright mobility, but many everyday activities in Indian contexts such as sitting cross-legged, cooking on the floor (chulha), and participating in social or cultural practices require a high degree of flexibility and comfort in seated postures. Users often find it difficult or uncomfortable to sit on the floor with their prosthesis, leading them to remove it altogether. This creates inconvenience, reduces independence, and limits participation in routine domestic and social activities."

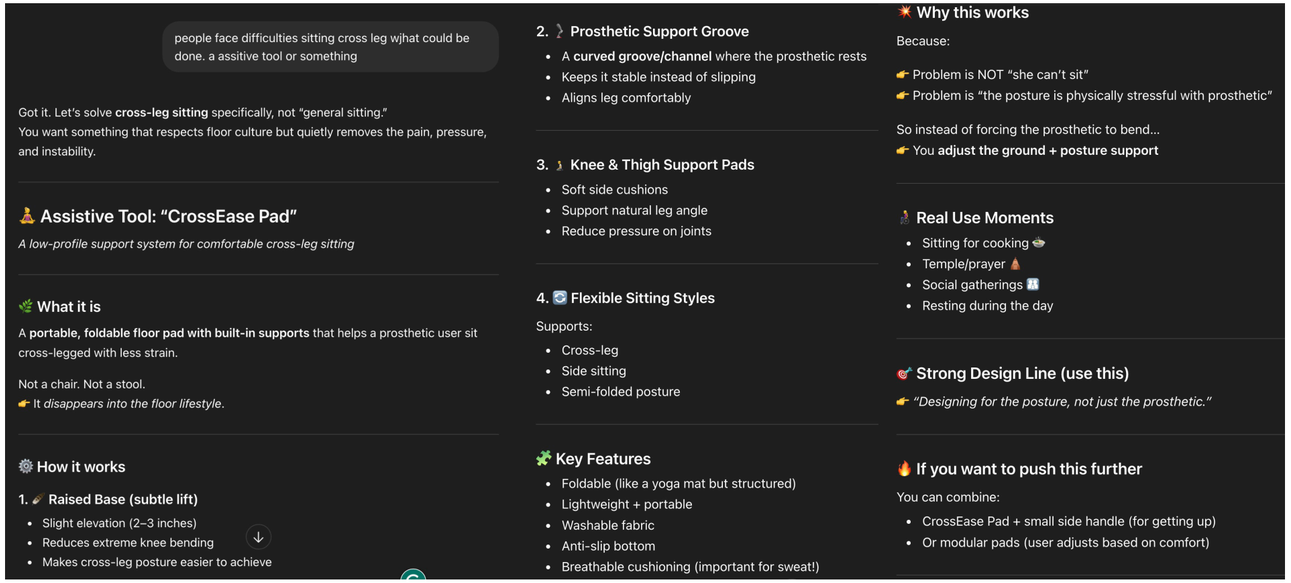
Evidence from interviews -During field observations and conversations, multiple users reported that they remove their prosthetic before sitting on the floor, as the device does not allow comfortable bending or posture adjustment. This was especially evident in activities like chulha cooking, where long durations of floor sitting are required. Users also mentioned discomfort, imbalance, and difficulty in maintaining posture while seated with the prosthetic.

Link to Ideas Phase 4:

Sol next page ->

CrossEase Pad

Ideation



people face difficulties sitting cross leg wjhat could be done. a assitive tool or something

Got it. Let's solve **cross-leg sitting** specifically, not "general sitting."
You want something that respects floor culture but quietly removes the pain, pressure, and instability.

🧘 Assistive Tool: "CrossEase Pad"

A low-profile support system for comfortable cross-leg sitting

🌿 What it is

A **portable, foldable floor pad with built-in supports** that helps a prosthetic user sit cross-legged with less strain.

Not a chair. Not a stool.
👉 It *disappears into the floor lifestyle.*

⚙️ How it works

1. 🪶 Raised Base (subtle lift)

- Slight elevation (2–3 inches)
- Reduces extreme knee bending
- Makes cross-leg posture easier to achieve

2. 🦿 Prosthetic Support Groove

- A **curved groove/channel** where the prosthetic rests
- Keeps it stable instead of slipping
- Aligns leg comfortably

3. 🦵 Knee & Thigh Support Pads

- Soft side cushions
- Support natural leg angle
- Reduce pressure on joints

4. 🔄 Flexible Sitting Styles

Supports:

- Cross-leg
- Side sitting
- Semi-folded posture

🧩 Key Features

- Foldable (like a yoga mat but structured)
- Lightweight + portable
- Washable fabric
- Anti-slip bottom
- Breathable cushioning (important for sweat!)

💥 Why this works

Because:

👉 Problem is NOT "she can't sit"
👉 Problem is "the posture is physically stressful with prosthetic"

So instead of forcing the prosthetic to bend...
👉 You **adjust the ground + posture support**

🧍 Real Use Moments

- Sitting for cooking 🍲
- Temple/prayer 🛕
- Social gatherings 🚻
- Resting during the day

🎯 Strong Design Line (use this)

👉 *"Designing for the posture, not just the prosthetic."*

🔥 If you want to push this further

You can combine:

- CrossEase Pad + small side handle (for getting up)
- Or modular pads (user adjusts based on comfort)

We started off with our problem statement regarding people being unable to sit on the floor without taking off their prosthesis and brainstormed solutions together with assistance from ai to ideate on an assistive tool that would enable people to be able to sit cross legged on the floor.

Concept

After coming up with our idea we worled on conceptualising the final concept for our solution- a tool that would enable people to sit on the floor without needing to take their prosthetic off. this product was mainly aimed at women who needed to sit on the floor to cook food on a chulha (traditional cooking stove) and sit inside religious places like temples.

Link to Ideas Phase 4:

Opportunity area -8

"Mobility aids like walkers are primarily designed to support standing and walking, but they do not adequately address the need for resting, sitting, or transitioning into lower postures, which are essential in many everyday activities. This creates a gap where users, especially those with limited mobility, experience fatigue and lack access to immediate seating or support in environments where chairs are not available."

Evidence from interviews -During observations and discussions, it was noted that users often relied heavily on walkers for support but had no integrated solution for resting or sitting, especially during prolonged use or while waiting in public spaces. This led to fatigue and dependence on external infrastructure.

Link to Ideas Phase 4:

Sol next page ->

Walker seat

Ideation

Another idea I have for my prosthetics and orthotics design project is a foldable walker with a vacuum pad-like mechanism on the bottom that the user can use to keep the walker in place so that its easier to hold as support for actions like squatting, what do you think?

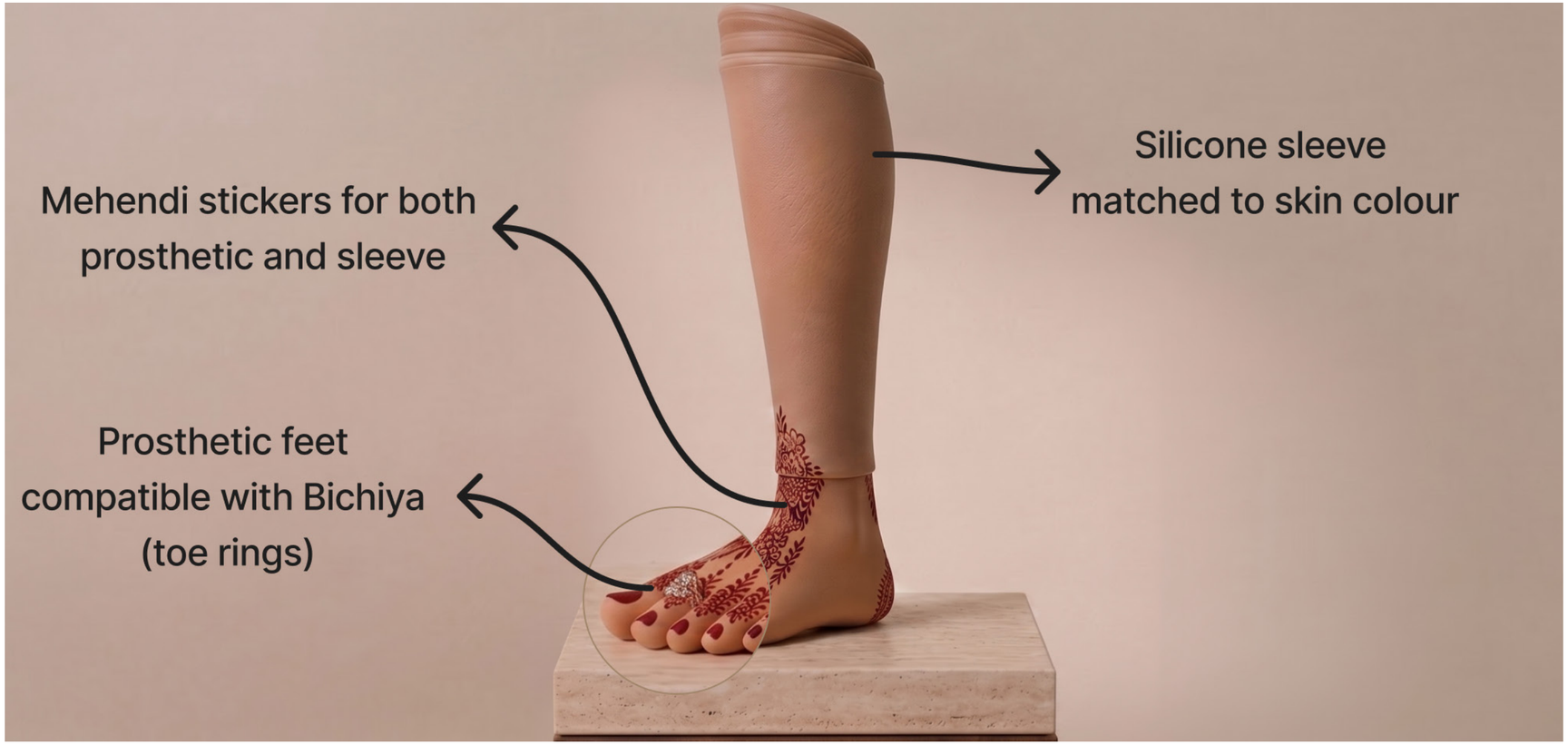
this is another idea that I had, can this become an attatchment for the walker so that the person can use it like a chair of sorts?

We wanted to explore the possiblity of making meaningful changes to pre existing mobility devices like walkers and came up with the idea of adding functionality to it by adding a suction pad to it so that it can be better used as support to hold during actions like squatting but very quicly realised during the brainstorming process that this version of the product would be more impractical since it would only work on certain surfaces, realising this, we pivoted to combining our last idea about the cross ease pad with this idea to be able to turn the walker into a foldable, portable seat.

Concept

Combining a mobility walker with our cross ease support pad we got to a concept of a foldable portable seat but with further research we realised that this product alread exists in other ways and forms in the current market due to which we decided to not iterate further on this idea.

How the prosthetic with our ideas might look



Link to Ideas Phase 4:

Opportunity area -9

"Prosthetic service delivery systems are largely centralised and event-based, requiring users to travel long distances to access assessment, fitting, and support. This model places a significant burden on users especially women and children who may face limitations due to mobility, finances, or social constraints. Additionally, the system lacks continuity of care, with minimal follow-up, guidance, or support once the prosthetic is provided. This creates a gap where access exists, but sustained usage, adaptation, and integration into daily life remain unsupported."

Evidence from interviews/ Secondary research -From field visits and ecosystem mapping, it was observed that most users discover services through informal networks like word-of-mouth. Secondary research reveals significant gender and geographic disparities in prosthetic access and outcomes in India.Data indicates strong gender and geographic disparities in prosthetic access in India. Of approximately 424,000 amputees, nearly 80% are from rural areas, with rural males forming 63%, while urban females account for only 4%, showing low visibility and access for women. Despite men being the majority, women have significantly lower success rates in prosthetic fitting (42.9% vs 68.6% for men) and are more likely to experience social isolation. Studies also show that women report higher dissatisfaction with fit, appearance, and limited footwear options, along with greater emotional and psychological challenges post-amputation due to societal pressures. There is also a clear demand for affordable, comfortable, and durable prosthetic solutions with reduced delays in fitting. These trends were reflected in our field visit at BMVSS, where out of nearly 50 users, only 2–3 were women, reinforcing the gap between need and actual access and highlighting how prosthetic care in India is shaped by gender, geography, and systemic barriers.

Link to Ideas Phase 4:

Sol next page ->

Service design

Care that reaches you, where you are

User understanding

Secondary Research

Synthesis

Defining the opportunity area

Current System understanding

Service design

Blueprint

Link to Secondary Research Excel :

Link to PDF Secondary Research :

Details in prototype stage

Service design

Care that reaches you, where you are

Designing Dignity Service blueprints

Service Design - make prosthetics and orthosis Available for all especially women and children

PHASE	Asha Worker goes to the households	Gives Beneficiary lists to GP/GS and PHC	Asha worker tries to take the careseeker to Jaipur foot(manual)	If care seeker cannot go , a mobile transportation is done (3...	If there is no jaipur foot center nearby, PHC/Gp requests for a camp
PHYSICAL EVIDENCE	**Asha Travel- van , car, bike, ASHA register / notebook, need assessment form**	**Paper beneficiary list Registers / handwritten records, stamps , forms**	**Travel, Bags, documents, Physical effort**	**Van/ vehicle , Medical transport setups, scanning equipment**	**Temporary camp setup Tools, materials Space, Queues, crowd**
CUSTOMER ACTIONS	Care Seeker • Receives ASHA at home • Shares condition (injury, amputation, difficulty walking) • Expresses needs or struggles • May or may not know about prosthetics • Asks questions hesitantly • Often influenced by family members	• Passive role (user is not actively involved) • Waits for updates • Relies on ASHA / local system • Often unaware of what is happening/ updates	• Decides whether to travel • Family support • Faces fatigue, discomfort	• Adjusts to schedule • Limited control over timing	• Waits for camp announcement • Travels locally • Gets assessed and fitted • Waits in queues
LINE OF INTERACTION					
FRONTSTAGE ACTIONS	• ASHA observes patient condition • Asks basic health related questions anf family related support • Gives verbal suggestions • Builds trust through familiarity	• ASHA submits names • GP/PHC acknowledges list • Officials may verify or ignore • No direct communication with user	• ASHA may guide or accompany • Provides verbal instructions	• P&O required for 3d scan • Coordinator to arrange to and fro of vehcile • ASHA informs users • Staff assists boarding	• Doctors + technicians interact • Measurement + fitting • Instructions given
LINE OF VISIBILITY					
BACKSTAGE ACTIONS	• ASHA notes potential beneficiary • Can help with pwd cards • Takes the list to GP/GS	• List aggregation • Basic filtering (sometimes) • Prioritization (sometimes) • No tracking system as such • Data may get lost or delayed	• Transport coordination • Arrangement , bookings	• Transport coordination • Route planning • Coordination between local bodies + Jaipur Foot • Capacity planning	• Camp planning • Resource allocation • Staff deployment • Bulk processing
LINE OF INTERNAL INTERACTION					
SUPPORT PROCESSES	• Government ASHA network • General health training	• Government administrative system • Paper-based workflows • No digital integration • No prosthetic-specific pipeline	• Informal travel planning • Family dependency	• P&O required • Funds • Ad-hoc logistics	• Government approvals • Donor funding
Possible Key Issues	• Awareness depends on ASHA knowledge • No systematized identification • Highly inconsistent outcomes	• Data flow is slow and unreliable • No feedback loop to user • No ownership of next step	• Physical burden • coordination • accessibility	• P&O required • Delivery and changes • might not be done in one day	• Episodic (not continuous) • No follow-up after camp - might need to go to nearby centers • Overcrowded, rushed

Link to Blueprint :

Link to Blueprint :

More Details in prototype stage

Faculty and peers review

We took feedback from faculty members and peers throughout the process to review our ideas, gain different perspectives, and refine our direction. This helped us critically evaluate our concepts and make more informed design decisions moving forward.

Prototype

Final Concepts(2)

How to make services available for women and children?

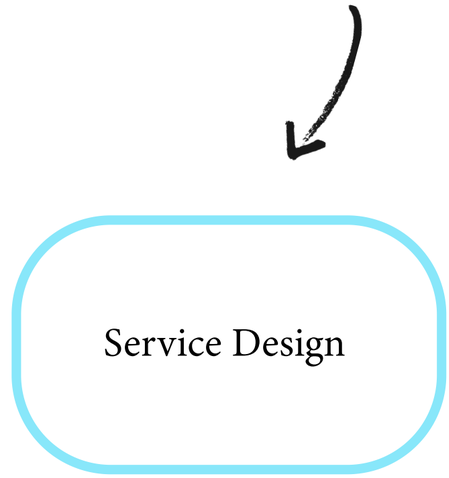
Service Design

How to enhance the experience of women with prosthetics (personalization)?

Chappal Lock

Part 1/2

How to make services available for women and children?



Brief of Service Design model

The proposed service model reimagines prosthetic care as a decentralised and continuous system, moving away from the current one-time, center-based approach. It builds on existing community networks such as ASHA workers, Primary Health Centres (PHCs), and local governance bodies to identify potential users early and create a more structured pathway to care. Instead of relying solely on individuals to travel long distances to centralized facilities, the model also introduces assisted access through mobility support, coordinated transportation, and localised interventions such as camps(no center available) and mobile units(center available). The inclusion of technologies like 3D scanning further enables preliminary assessments and data collection to happen closer to the user, reducing delays and repeated travel.

The model also emphasises better coordination between stakeholders by introducing clearer communication flows and basic tracking mechanisms, ensuring that users are not lost in the system after initial identification. This helps streamline the journey from awareness to fitment, making the process more transparent and accessible.

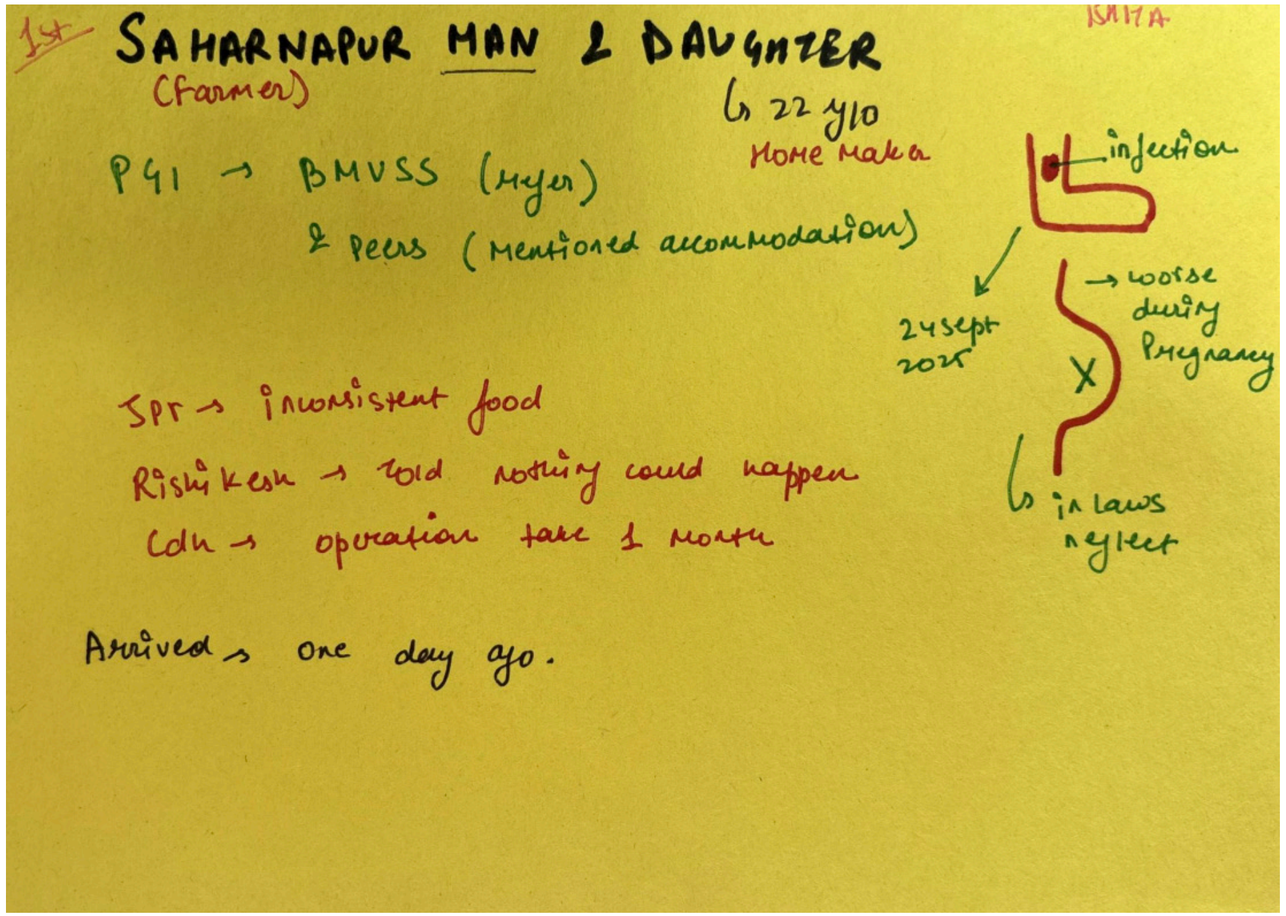
Brief of Service Design model

A key focus of the model is on post-fit continuity, addressing one of the most critical gaps in the existing system. It proposes home-based follow-ups, minor adjustments, and contextual training that helps users adapt the prosthetic to their daily lives, including activities such as sitting, walking on different terrains, and managing hygiene. **This is particularly important for women, who often face additional social and mobility constraints and whose needs are not adequately addressed in current systems.**

By integrating community-level support with centralized expertise, the model shifts prosthetic care from an episodic service to an ongoing support system. It aims to improve not just access, but also long-term usability, comfort, and user confidence, ensuring that prosthetic care extends beyond fitment and becomes embedded in the user's everyday life.

User Understanding real life Scenario

[Link to ALL Women interviews :]



A farmer from Saharanpur arrived in Jaipur with his 22-year-old daughter. Her condition had begun as an infection around September 2025, but during her pregnancy it worsened rapidly, eventually affecting her ankle severely. By the time they reached proper medical attention, the situation had already deteriorated(in laws neglected care). The baby did not survive, and the infection had spread further

They had travelled across cities Rishikesh, Chandigarh searching for treatment, but each place brought new challenges. In Chandigarh, they were told the process could take a month. For a family dependent on farming, both time and money were limited resources. They finally came to Jaipur after someone they knew mentioned the place, mainly because accommodation might be available. Otherwise, they had no clear direction on where to go or which hospital to trust.

[Link to ALL Women interviews :]

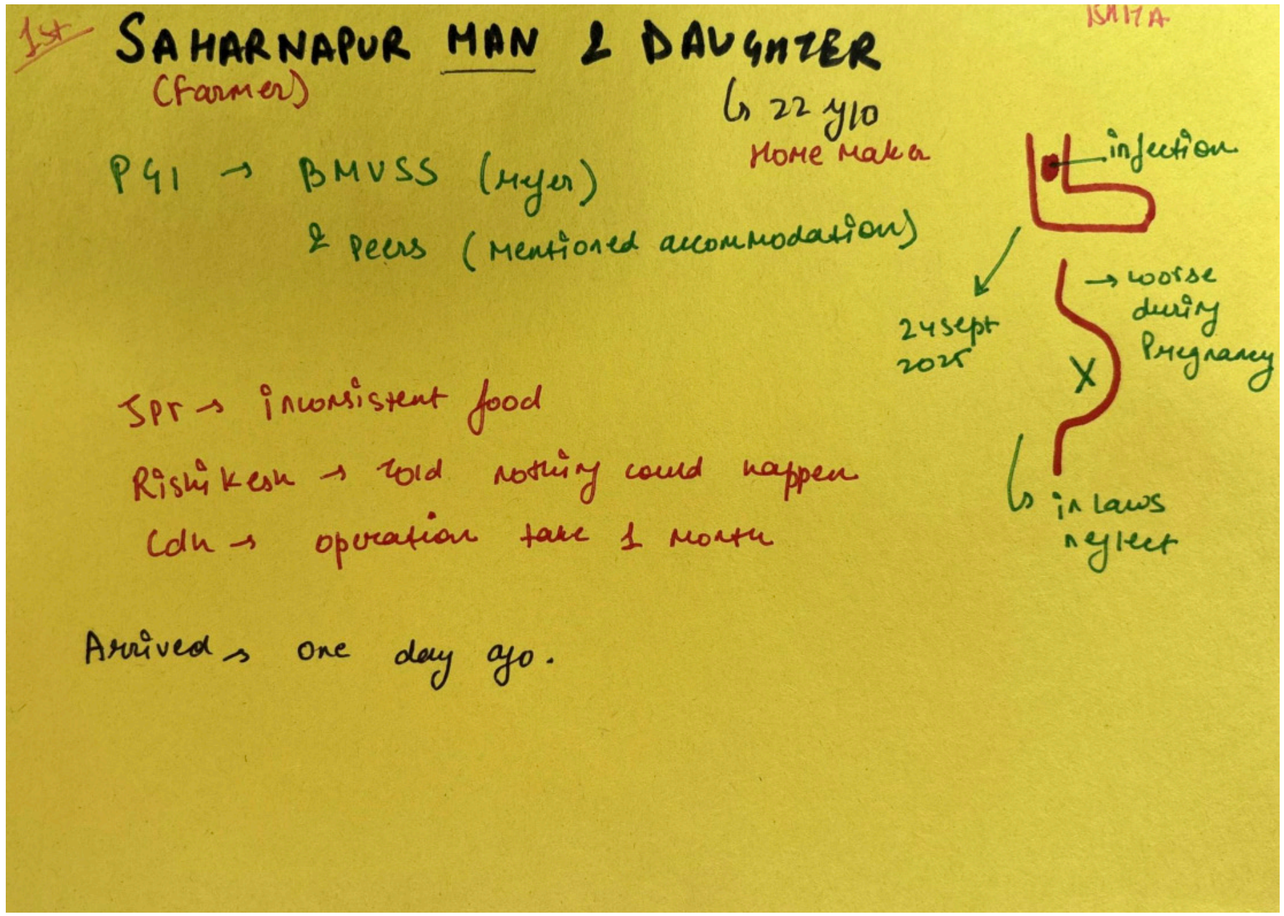
User Understanding real life Scenario

[Link to ALL Women interviews :]

The daughter, who had been a homemaker after her marriage two years ago, was now unable to carry out even basic daily activities. It was also shared that her condition worsened because her in-laws did not get her infection checked earlier, which added another layer of emotional strain.

In Jaipur, they sometimes managed to get free food, but it was inconsistent. Each step of the process travel, treatment decisions, waiting was physically exhausting, financially draining, and emotionally overwhelming. They were not just dealing with a medical condition, but also with loss, confusion, and the pressure of not knowing what would happen next.

Their story reflects a reality faced by many where access to the right care is not straightforward, and the journey to treatment becomes as difficult as the condition itself.



[Link to ALL Women interviews :]

Service Design Secondary Research

IIT Guwahati creates affordable prosthetic leg for Indians that allows deep squatting, cross-legged sitting	IIT Guwahati creates affordable prosthetic leg for Indians that allows deep squatting, cross-legged sitting - India Today	
Focusing Research Efforts on the Unique Needs of Women Prosthesis Users	Focusing Research Efforts on the Unique Needs of Women Prosthesis Users	1. Women with limb loss experience greater dissatisfaction with prosthetic prosthetics solutions to accommodate the desire of women to wear differ footwear is important to attire and hence community participation. Despit research, which limits available information to guide the rehabilitation pro 2. Women represent a growing portion of the population with lower-limb are limited prosthetics solutions to accommodate the needs of women wit optimal mobility outcomes
Psychological Flexibility in People with Lower Limb Amputation among Males & Females in Indian Population	Psychological Flexibility in People with Lower Limb Amputation among Males & Females in Indian Population	The result of the studies shows that there was a gender difference in conte females with lower limb amputation are psychologically less flexible than r psychological inflexibility is a major problem post amputation more comm psychological disturbances which can be a painful memory leading to a les to the male dominant society therefore it will be helpful to understand the them a better & more satisfactory life.
Identifying and Acknowledging a Sex Gap in Lower-Limb Prosthetics Research: A Review	Identifying and Acknowledging a Sex Gap in Lower-Limb Prosthetics Research: A Review	
Depression and anxiety symptoms after lower limbamputation: the rise and fall	https://journals.sagepub.com/doi/epdf/10.1177/0269215508094710	
Quality of life of prosthetic and orthotic users in South India: a cross-sectional study	https://link.springer.com/article/10.1186/s12955-019-1116-y	
Examining the needs of affordable upper limb prosthetic users in India: A questionnairebased survey	https://sci-hub.kvnp.top/10.3233/TAD-160448	A questionnaire-based survey has been carried out on UL prostheses users highlight the demand for low-cost prosthetic arms that can provide functio sockets and harness systems that are both cost-effective and durable. The durability. Additionally, glove durability is a major area of concern. Further, prosthetic use and enable a much greater rate of acceptance. Accurately c into appropriate solutions. Future work in this project will propose affordab prosthetic rehabilitation in this developing world setting.
National Health Mission	https://nhm.gov.in/index1.php%3Flang%3D1%26level%3D2%26sublinkid%3D150%26lid%3D226	This shows the role of Asha Workers
Beneficiary Report	https://egramswaraj.gov.in/beneficiaryReport.do	
Gram Panchayat	https://grammanchitra.gov.in/gm4MVC	
Food security act	https://nfsa.gov.in/portal/nfsa-act	
Drive link multiple papers	https://drive.google.com/drive/folders/1pIU4p7wOwtdYxBq9piqOKZxGGAYzQAM6	MULTIPLE PDFS

Link to Secondary Research (FULL ACCESS) :

Link to PDF'S secondary Research :

Secondary research was conducted to understand how the existing grassroots healthcare system in India functions, particularly the roles of ASHA workers, Panchayats, and Primary Health Centres (PHCs). ASHA workers are a key component of India's public health system, introduced under the National Rural Health Mission to act as a link between rural communities and formal healthcare services. They are typically women from the same village and serve as the first point of contact for healthcare needs, especially for populations that find it difficult to access hospitals or clinics.

Their responsibilities include conducting household visits, identifying patients, spreading awareness, and facilitating access to services like PHCs and government schemes. They also play a crucial role in referrals, data collection, and mobilising communities, working closely with Panchayats and local health committees.

Service Design Secondary Research

At a systemic level, PHCs function as the primary healthcare nodes, where patients are referred for diagnosis and treatment, forming an important part of the rural healthcare infrastructure.

Research also highlights that ASHA workers are considered the backbone of community-level healthcare, effectively bridging gaps between underserved populations and institutional systems, especially in rural areas. However, the system remains largely paper-based, fragmented, and dependent on individual workers, with challenges in coordination, tracking, and continuity of care.

Understanding this structure helped in identifying how prosthetic services currently rely on informal networks, referrals, and centralized systems, and where opportunities exist to build a more connected, accessible, and continuous service model.

IIT Guwahati creates affordable prosthetic leg for Indians that allows deep squatting, cross-legged sitting	IIT Guwahati creates affordable prosthetic leg for Indians that allows deep squatting, cross-legged sitting - India Today	
Focusing Research Efforts on the Unique Needs of Women Prosthesis Users	Focusing Research Efforts on the Unique Needs of Women Prosthesis Users	1. Women with limb loss experience greater dissatisfaction with prosthetic prosthetics solutions to accommodate the desire of women to wear differ footwear is important to attire and hence community participation. Despit research, which limits available information to guide the rehabilitation prc 2. Women represent a growing portion of the population with lower-limb are limited prosthetics solutions to accommodate the needs of women wi optimal mobility outcomes
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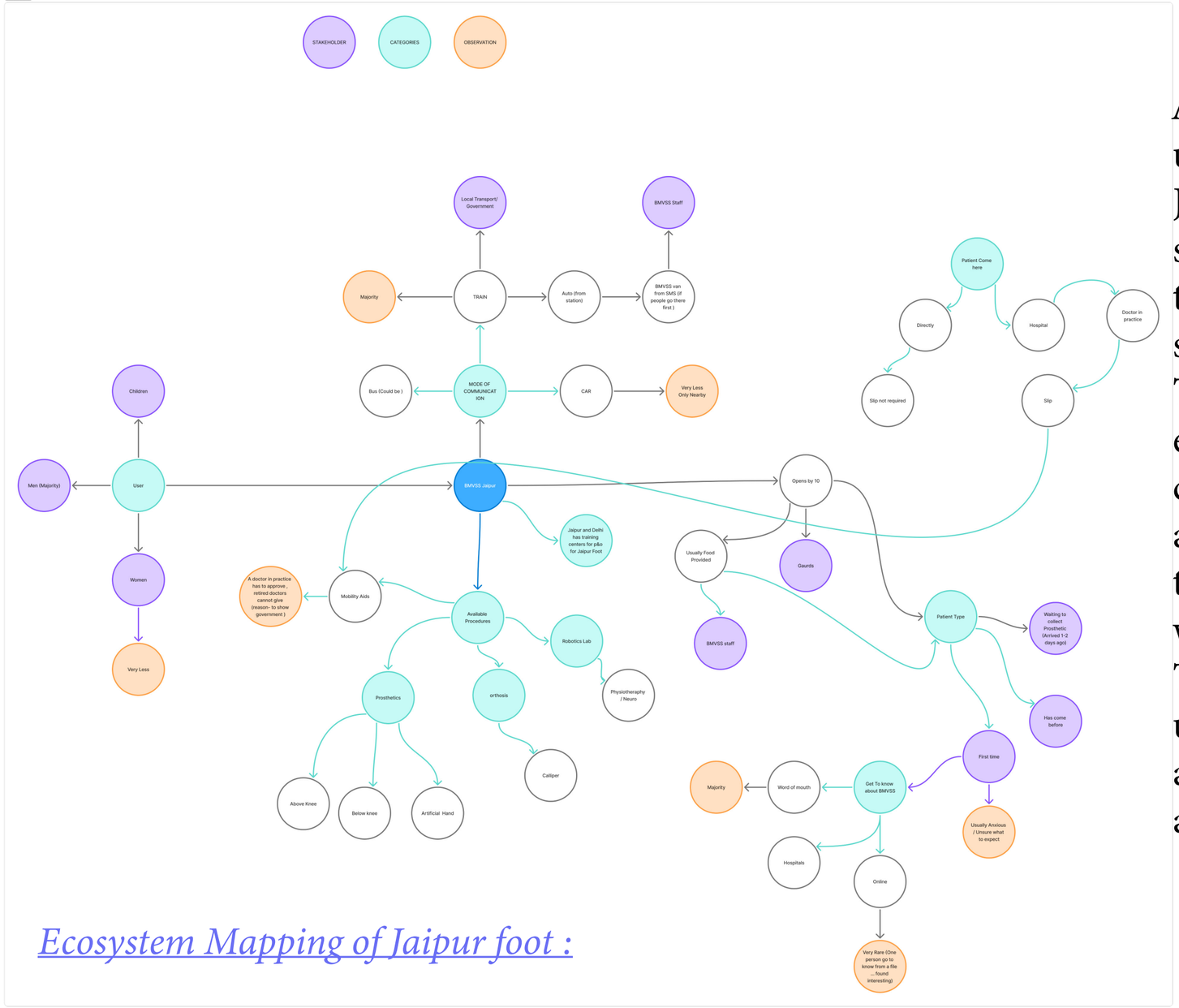
National Health Mission	https://nhm.gov.in/index1.php%3Flang%3D1%26level%3D2%26sublinkid%3D150%26lid%3D226	This shows the role of Asha Workers
Beneficiary Report	https://egramswaraj.gov.in/beneficiaryReport.do	
Gram Panchayat	https://grammanchitra.gov.in/gm4MVC	
Food security act	https://nfsa.gov.in/portal/nfsa-act	
Drive link multiple papers	https://drive.google.com/drive/folders/1pIU4p7wOwtdYxBq9piqOKZxGGAYzQAM6	MULTIPLE PDFS

Link to Secondary Research (FULL ACCESS) :

Link to PDF'S secondary Research :

Ecosystem mapping of BMVSS(Jaipur Foot)



[Ecosystem Mapping of Jaipur foot :]

[Ecosystem Mapping of Jaipur foot :]

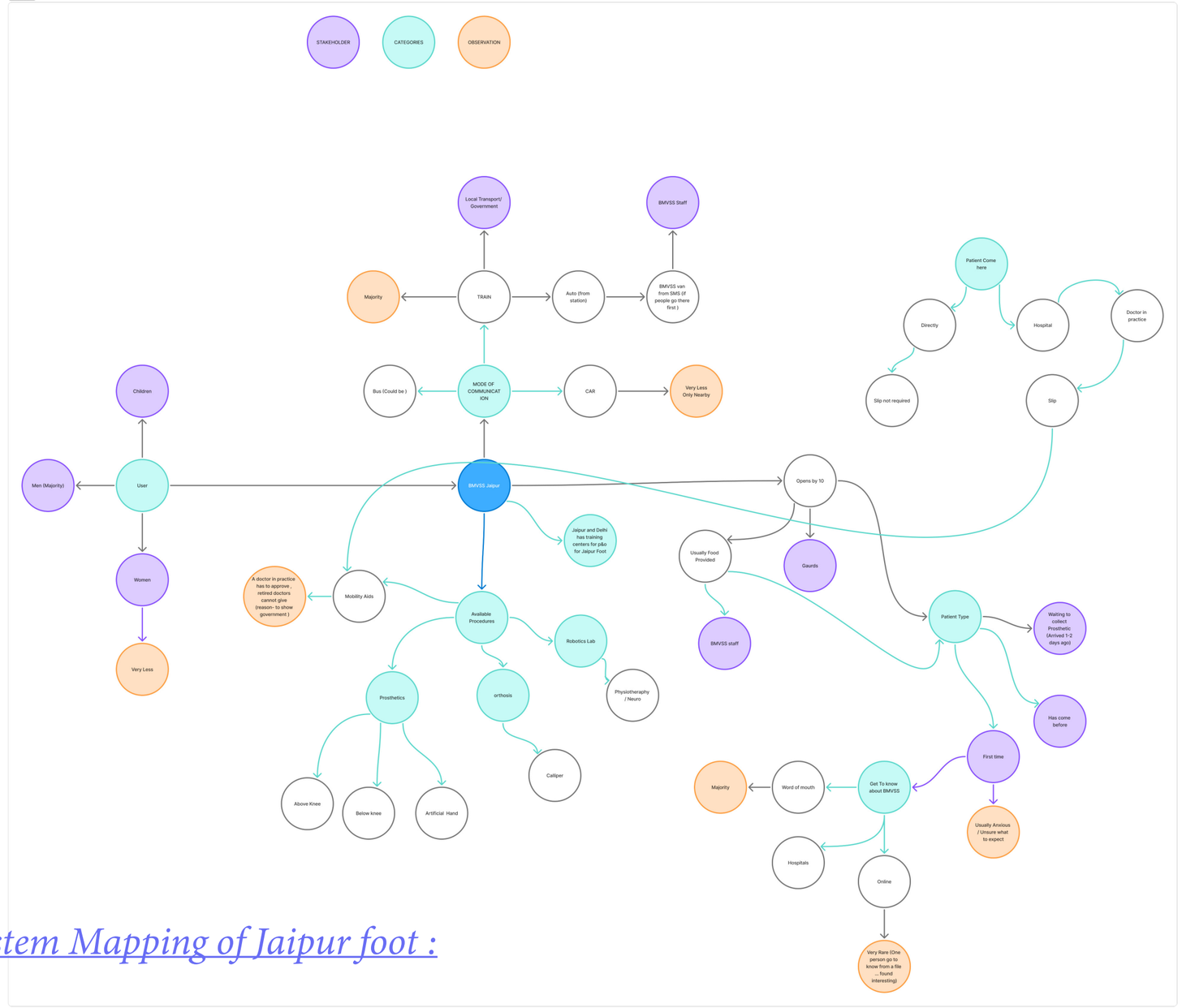
An ecosystem mapping of BMVSS was conducted to understand the existing working and structure of the Jaipur Foot system. This included identifying the key stakeholders involved, such as patients, families, technicians, doctors, administrative staff, and supporting networks like NGOs and donors.

The mapping helped in understanding how users enter the system, the flow of processes within the center, and how services like assessment, fabrication, and fitment are carried out. It also provided clarity on the roles of different actors and how they interact within the system.

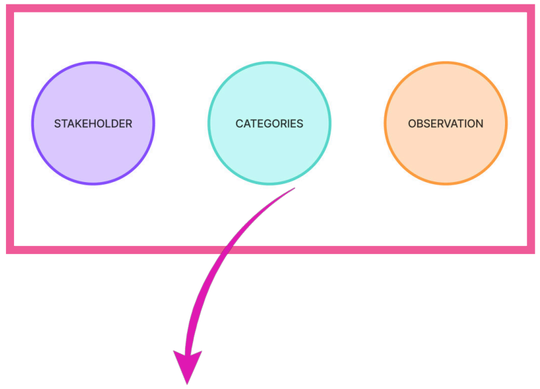
This exercise was primarily done to gain a clear understanding of how Jaipur Foot currently functions as a service system, forming a base for further analysis and intervention.

Ecosystem mapping of BMVSS(Jaipur Foot)

Section 1



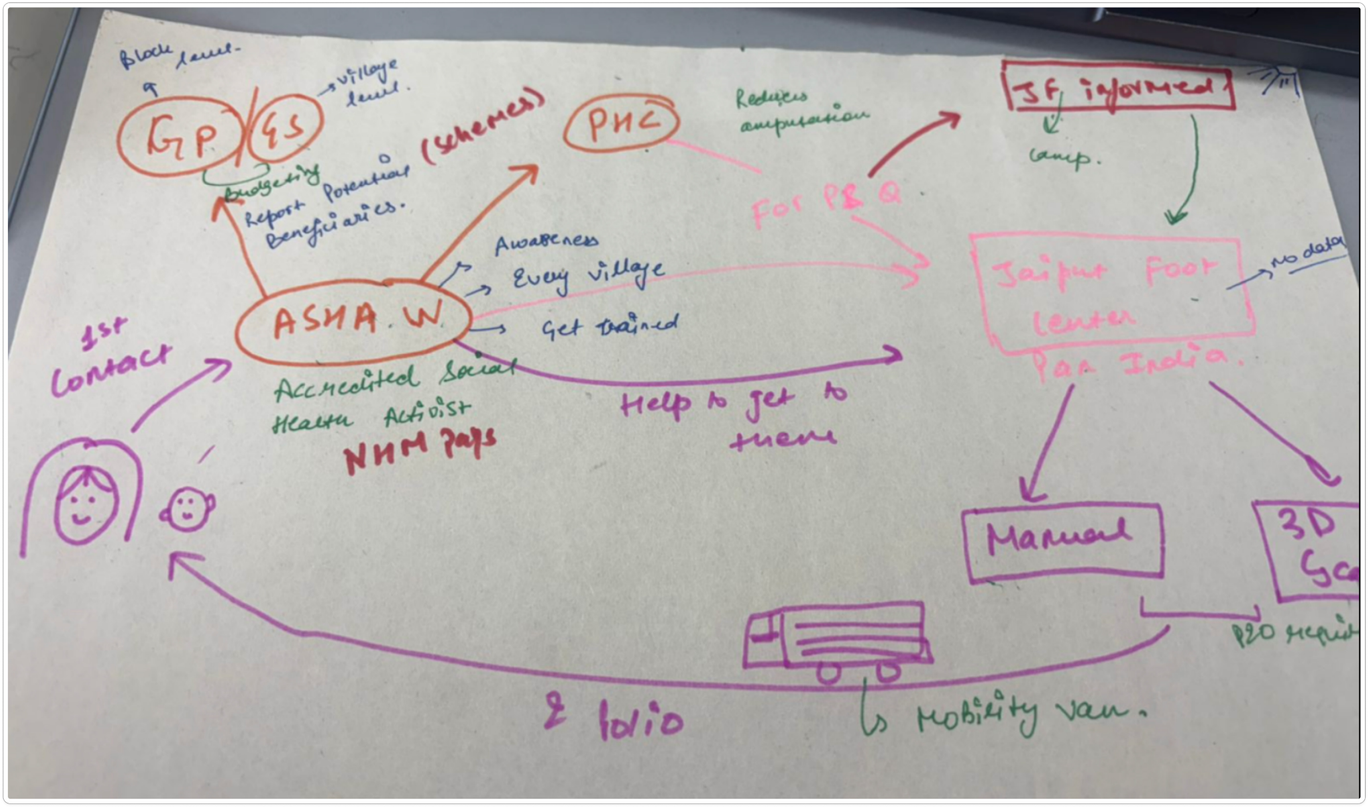
Ecosystem Mapping of Jaipur foot :



Ecosystem Mapping of Jaipur foot :

Proposed Service model

Link to Service Model mapping :

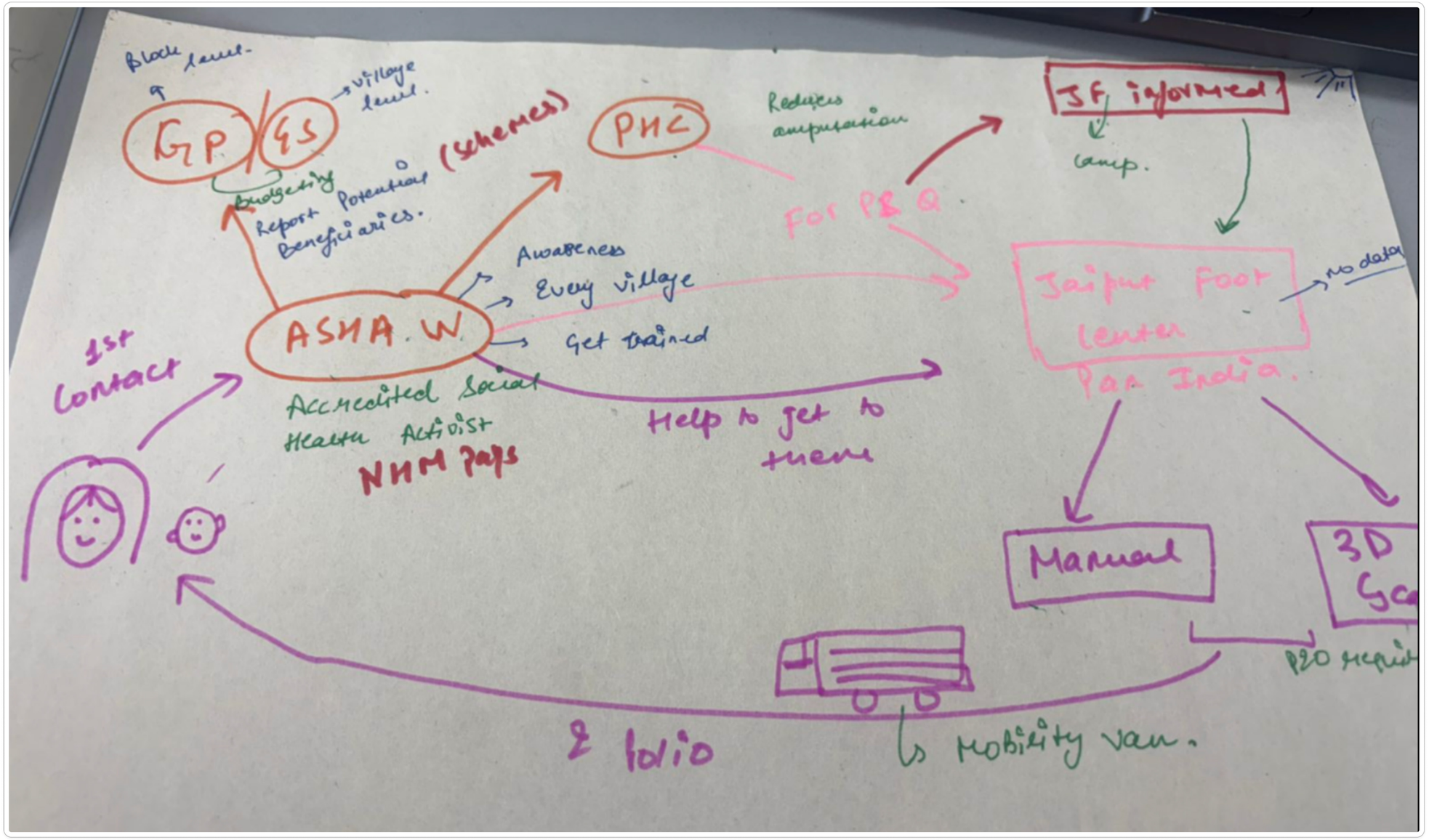


Based on insights from primary research, ecosystem mapping, and identified gaps in access and continuity, a proposed service model was developed to make prosthetic care more accessible, coordinated, and continuous. The model shifts from a purely centralized system to a more decentralised and community-linked approach, leveraging existing structures such as ASHA workers, Panchayats, and PHCs.

Link to Service Model mapping :

Proposed Service model

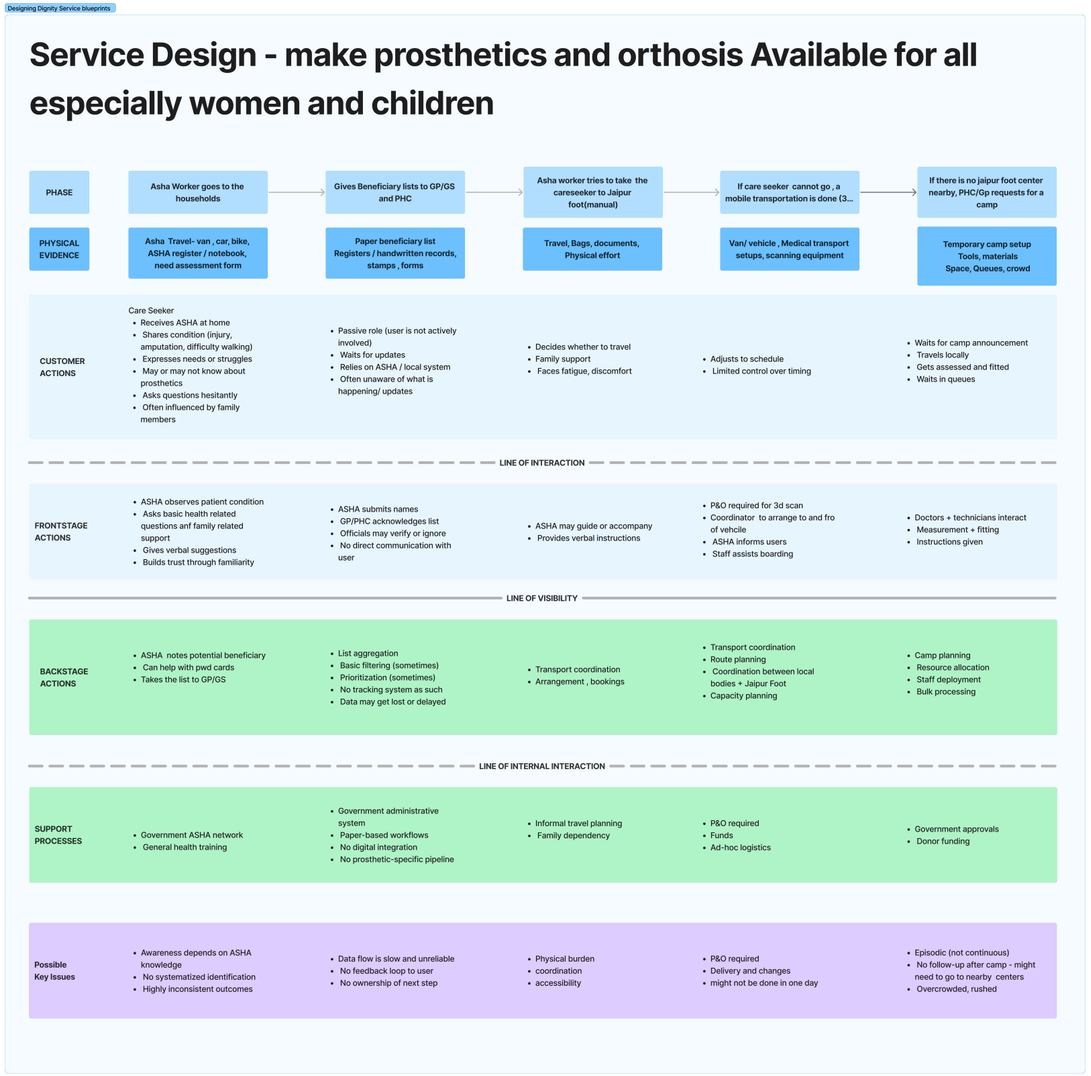
[Link to Service Model mapping :]



[Link to Service Model mapping :]

Service Blueprint

Link to Blueprint :



To better understand and structure the proposed service model, a service blueprint was created mapping the entire user journey across different stages. The blueprint includes layers such as physical evidence, customer actions, frontstage interactions, backstage processes, and support systems, providing a holistic view of how the service operates.

It also allowed us to map how different stakeholders like ASHA workers, PHCs, mobility services, and centralized centres interact within the system.

The blueprint served as a key tool to translate insights into a structured service flow, ensuring that the proposed model addresses gaps not just at a product level, but across the entire experience.

Link to Blueprint :

Service Blueprint

[Link to Blueprint :]

Designing Dignity Service blueprints

Service Design - make prosthetics and orthosis Available for all especially women and children

PHASE	Asha Worker goes to the households	Gives Beneficiary lists to GP/GS and PHC	Asha worker tries to take the careseeker to Jaipur foot(manual)	If care seeker cannot go , a mobile transportation is done (3...	If there is no jaipur foot center nearby, PHC/Gp requests for a camp
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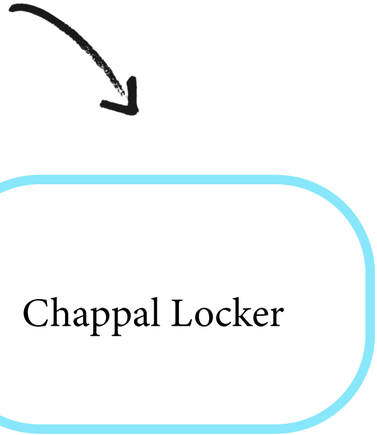
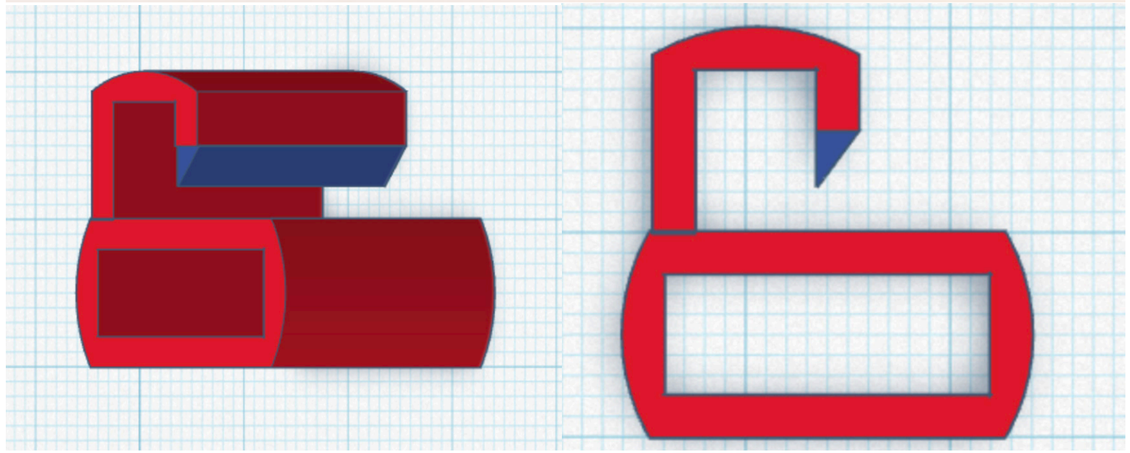
[Link to Blueprint :]

Part 2/2

How to enhance the experience of women with prosthetics (personalization)?

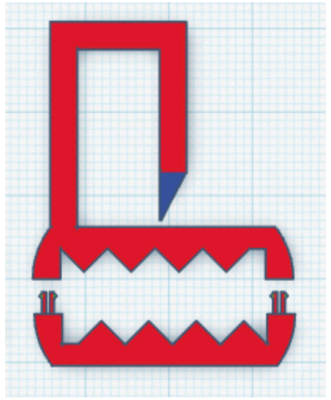


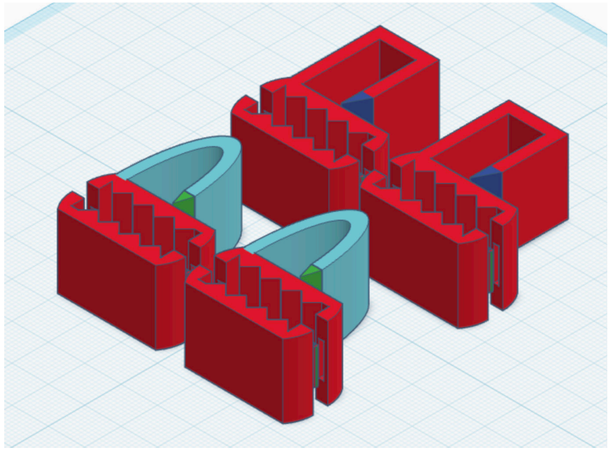


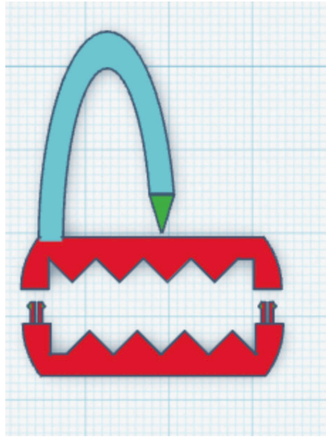
Prototyping phase

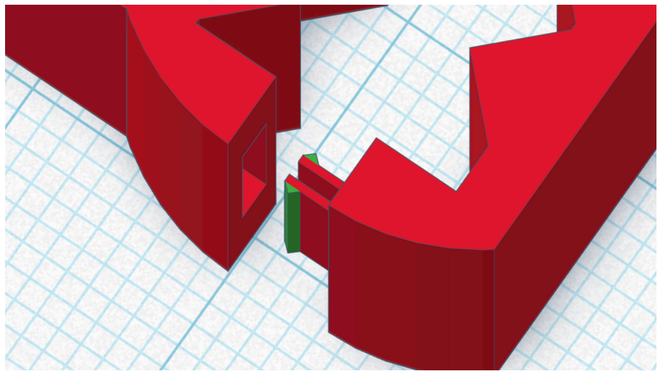
Out of all of these concepts and ideas, we decided to take the Chappal Lock and the universal footwear lock for juttis, mules etc. forward to the prototyping stage.

We decided to go with CAD modelling combined with 3D printing to create a prototype for the clip that will attatch to the chappal strap and have some kind of hook to hold our heel strap add-on.







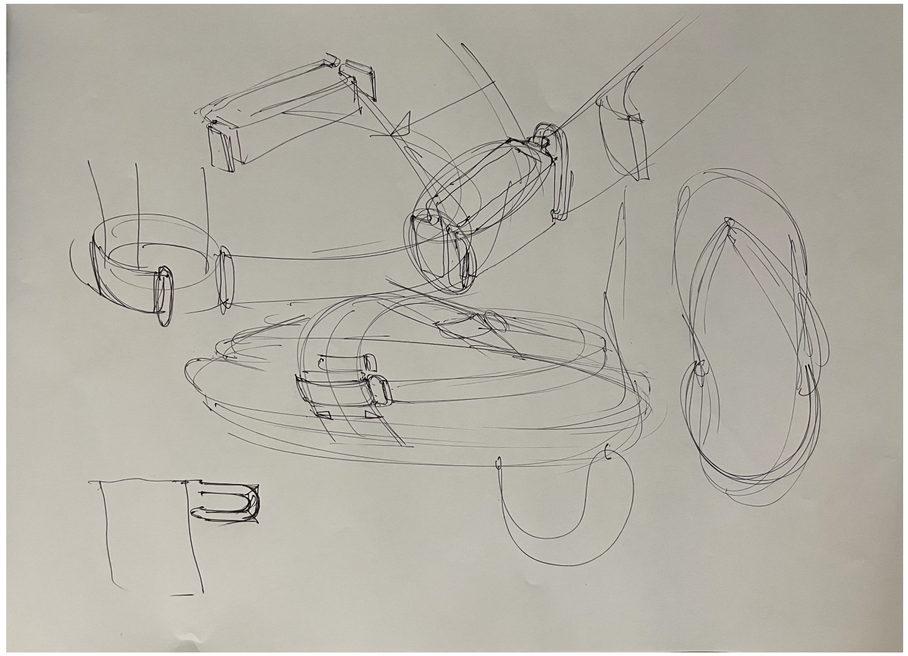


Prototype 1

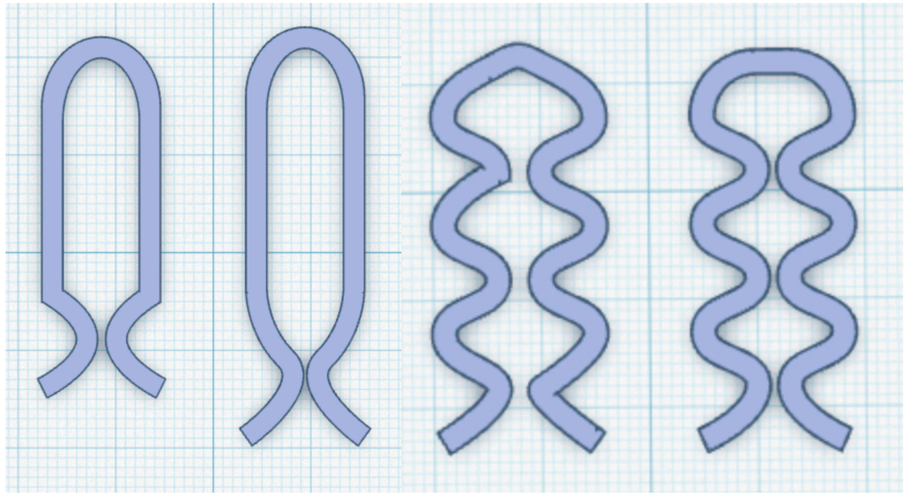
The first version of our prototype was very barebones, all we needed was a way to attatch out model to the chappal straps and a hook for our heel strap attatchment.

For the first part we decided to cut our model in 2 halves and then researched on mechanisms we can use to connect these two together before landing on our final solution of adding a snap-in joint to the parts so they can connect.

As for the hook initially we had 2 versions, the first one was much more anglular while the other one was curved. unfortunately due to time constraints we were never able to print out and test this prototype.



Product sketches made by Sukumaran S S

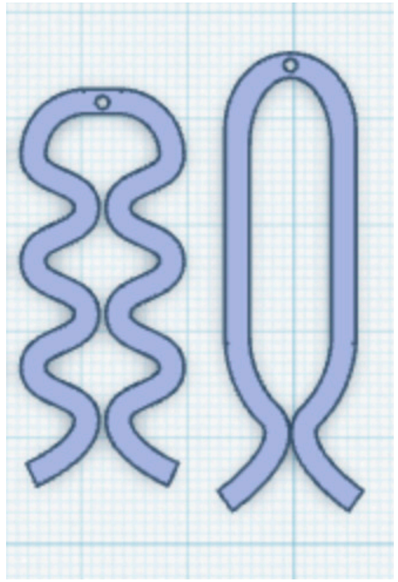


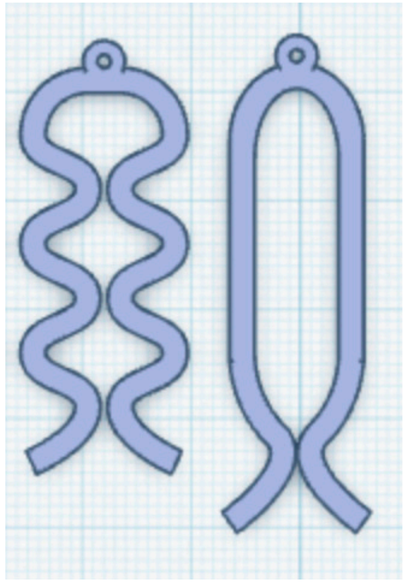
Iterating on shapes

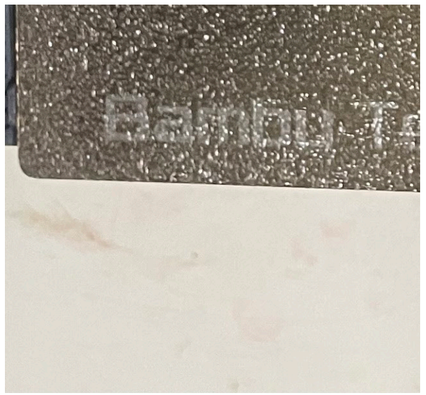
Prototype 2- conceptualization

We based this version of our prototype based off off our professor's feedback on our first prototype and the product sketches he created as part of this feedback. Instead of having a clip on mechanism and a hook like the last prototype this one was supposed to slide into the strap of a chappal and then be secured to it using a rubber band/O ring. Furthermore this design would have a hole in it instead of a hook and a metal hook would go in it for better structural integrity.

While iterating on the shape for this design in the CAD software we realized that this design has no grip with the strap of the chappal which could lead to it sliding up and down the strap. to tackle this problem we iterated on the next version of this prototype that had a ribbed design with grooves that would maintain grip with the chappal. Additionally a variation of this design could also work for the universal footwear person of this idea.





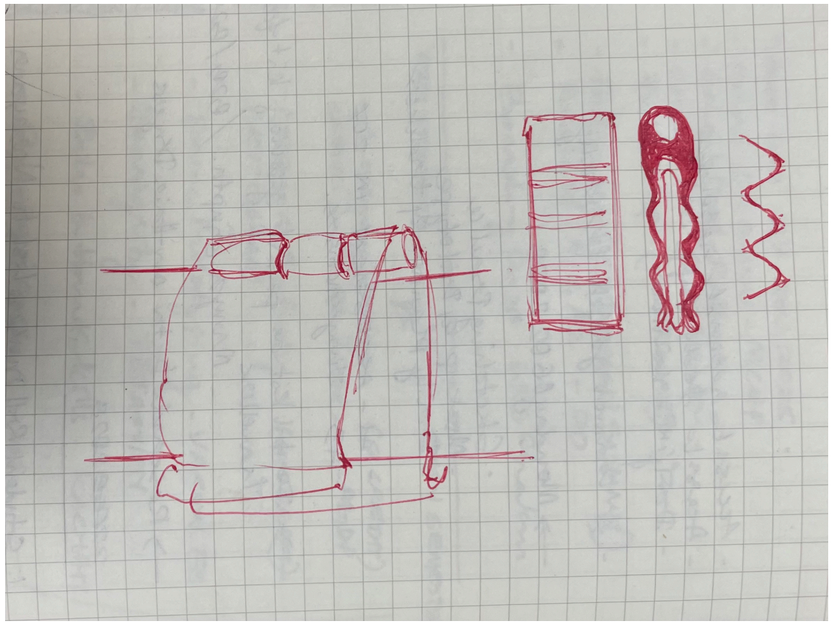


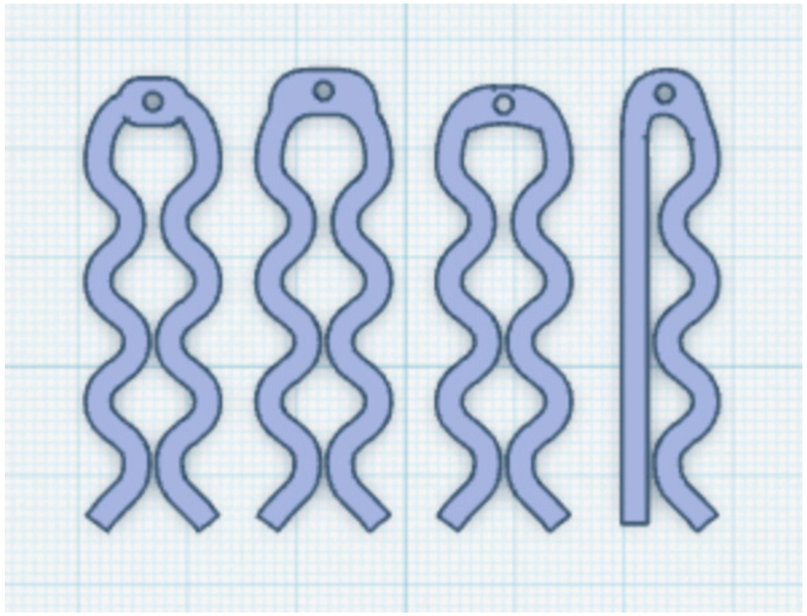
Prototype 2- fabrication

While we had finalized a shape for the next prototype on our cad software, we still needed to add a hole to so that we could attach a metal ring to it (in this case made using a paper clip) that would hold the strap.

To solve this problem we experimented with 2 solutions, first was to add a hole to the existing design itself and the second was to create a cylindrical extrusion to the clip with a hole to hold the metal ring.

Once we 3D printed the models only one of these implementation ended up working out. the first implementation (hole in the design) ended up being too fragile and was broken right off of the print bed, making the second design our winner by default.



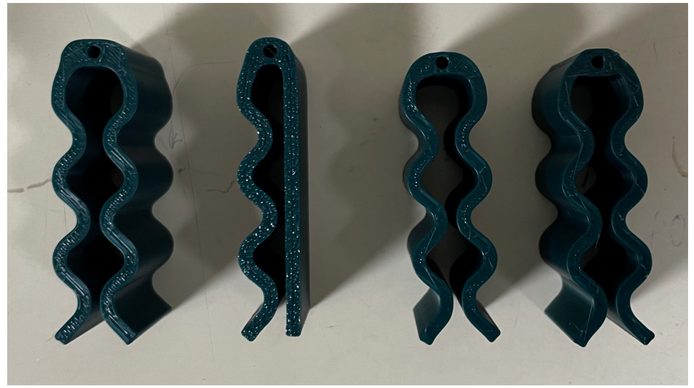


Prototype 3- conceptualization

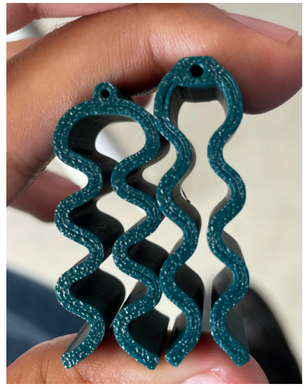
Just like last time, this prototype also started off as feedback and sketches from our professor, but the changes this time around are much more minor and iterative.

We retained the ribbed and grooved design from the last prototype while making it thinner and replaced the cylindrical extrusion for the metal loop with a thicker top part of the clip for better structural integrity. Additionally, we made more minor changes based on our user testing like increasing the gaps between the grooves closer to the top since that section has less flex making us struggle with inserting it into the straps of slippers. Additionally all of these concepts also had larger holes for the metal ring since we faced issues with an extremely tight fit last time around.

Prototype 3- fabrication & implementation



All iterations of the 3rd prototype



Prototype 2 VS 3

Current prototype/ proof of concept

This time around we got all of our prints with no issues at all. We had 4 different versions of this prototype, 3 of which were experimenting with different ways to make the top part thicker and one where one side was flat and the other was ribbed. This was done since we were also looking into implementing this product for people who aren't amputated but still face similar issues with footwear. the flat side would be the one in contact with their feet for better comfort.

After getting this clip design approved by our professor and peers, we created a proof of concept of this idea by using paper clips as our makeshift metal rings and an adjustable elastic strap attached to it.

The product held up well in our limited user testing and seems to be ready to be taken to the final stage where we want to use an elastic material like polypropylene for the clips, proper metal rings and a transparent adjustable strap going through the back of the foot.

Potential Output

CHAPPAL *LOCK SYSTEM*

ADDITIONAL HEEL RETENTION FOR STABILITY

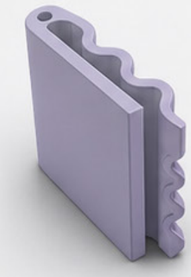
A modular, grip-on heel strap system that instantly upgrades any open-back footwear for improved stability and secure fit. Designed for prosthetic and orthotic users.

NO MODIFICATION
Clip-on design. Fits most footwear.

MORE STABILITY
Helps prevent heel slip and improves security.

LIGHTWEIGHT & DURABLE
PP clip and TPU strap for long-lasting everyday use.

COMPONENTS



CLIP (POLYPROPYLENE)

- Curved inner profile for a secure, anti-rotation fit
- Textured inner surface for better grip
- Durable, lightweight and impact resistant

STRAP (TRANSPARENT TPU)

- Frosted TPU for flexibility, strength and skin comfort
- Wider heel zone reduces friction and pressure
- Adjustable length for custom fit

SNAP HOOK (POLYPROPYLENE)

- Quick and easy to attach / detach
- Secure locking with audible click
- Smooth edges for skin safety

DETAILS

TEXTURED INNER GRIP
Prevents slipping and movement even during active use.

SECURE SNAP HOOK
One-handed operation with a firm locking click.

WIDER HEEL CONTACT
Distributes pressure evenly and improves long-term comfort.

MATERIALS

POLYPROPYLENE (PP)

- Lightweight
- High impact strength
- Good fatigue resistance
- Easy to clean
- Cost effective

TRANSPARENT TPU

- Flexible & soft
- Wear resistant
- Skin friendly
- Transparent frosted finish

HOW IT WORKS

1 CLIP ON
The polypropylene clip snaps securely onto the side wall of the chappal.

2 ATTACH STRAP
The strap's snap hook connects to the clip for a secure lock.

3 LOCKED IN
Adjust the strap for a snug, comfortable fit. Ready to go.

KEY FEATURES

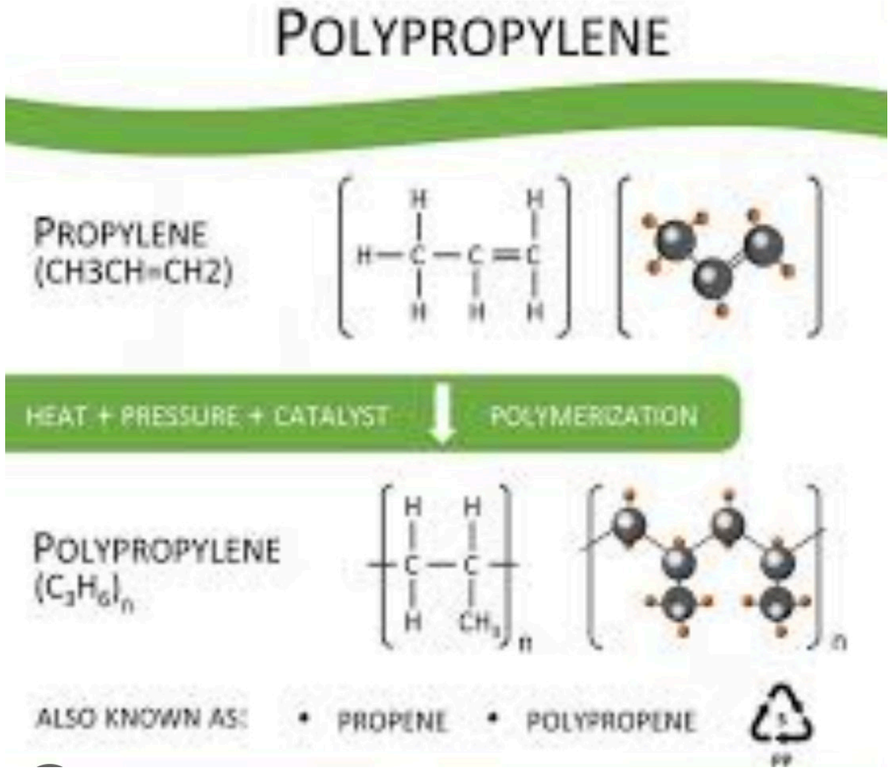
- **Universal Fit** — Works with most open-back footwear
- **Improved Stability** — Helps reduce heel lift and improves gait confidence
- **Modular & Detachable** — Easy to attach, remove or replace
- **Durable Materials** — Built for daily use and long-term durability
- **Comfort Focused** — Soft strap and smooth edges for all-day comfort
- **Lightweight Design** — Minimal weight addition for effortless use

COLOUR OPTIONS

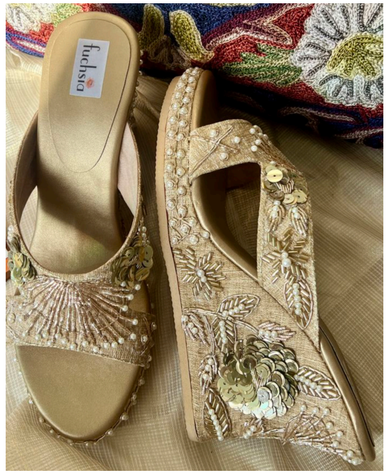
Simple. Secure. Stable.
Designed for better mobility, every day.

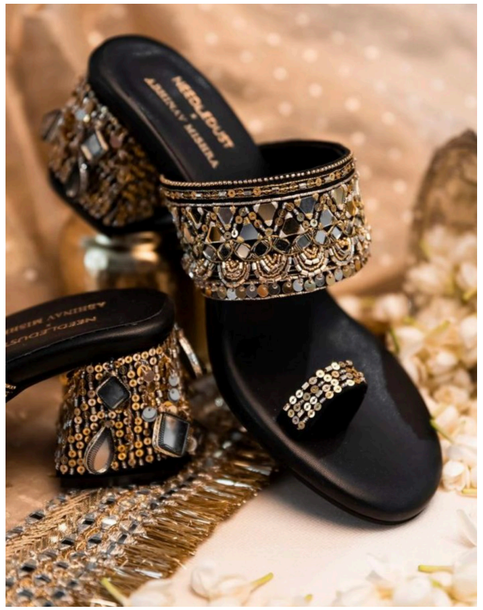
Materials to be used

polypropylene



Foot wear it can used on





many more....

Future Scope

Chappal Lock : The proposed solutions, both at the product and system level, have the potential to expand beyond their current scope and become more inclusive and scalable. The chappal lock concept can be adapted for a wider range of users, including elderly women and even able-bodied individuals who face issues with unstable footwear, making it a more universal solution rather than being limited to prosthetic users. It can also evolve into multiple variations in terms of design, materials, and colors, allowing users to choose based on personal preference and cultural context, while maintaining the same core functionality of providing stability and grip.

Service : At a system level, the service model can be scaled across regions by integrating more deeply with existing public health infrastructure such as ASHA workers, PHCs, and local governance bodies. It can further incorporate digital tools for tracking, coordination, and communication, improving efficiency and continuity of care. The model can also expand to include training for local workers, enabling them to provide basic support, follow-ups, and guidance at the community level.

Together, these interventions can evolve into a more comprehensive ecosystem that not only improves access to prosthetic care but also enhances long-term usability, independence, and inclusion across different user groups and contexts.

Links

Link to Ideas Phase 1 : exploration

Link to Ideas Phase 2 : categorization

Link to Ideas Phase 3 : tools execution

Link to Ideas Phase 4: Generated

Link to Secondary Research Excel :

Link to PDF Secondary Research :

Link to Blueprint :

Link to Secondary Research (FULL ACCESS) :

Link to PDF'S secondary Research :

Link to ALL Women interviews :

Ecosystem Mapping of Jaipur foot :

Link to Service Model mapping :

THANK YOU

Designing for all by designing for Niche